

Recap: 空间域中的图像增强

空间域图像增强

基于简单灰度变换的图像增强

基于直方图均衡和规定化的图像增强

基于多幅输入图像的图像增强



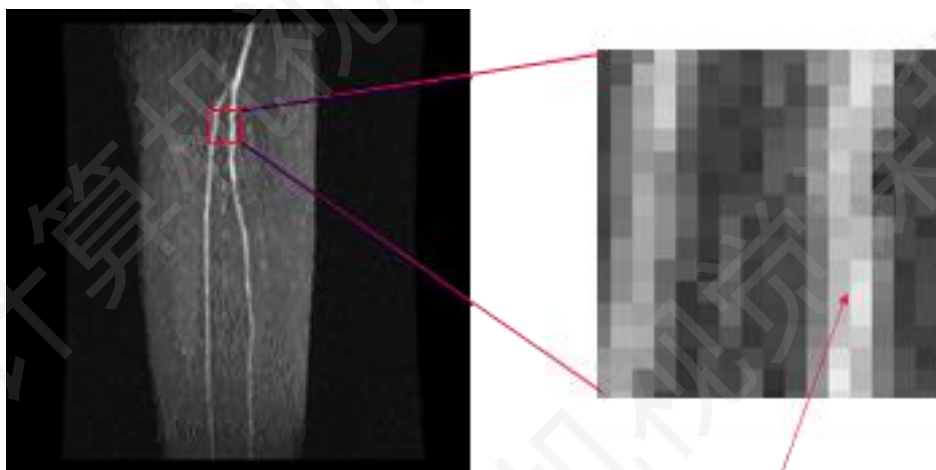
any more?

有什么共性问题??

没有考虑图像的**空间信息**（输出处理结果中像素的灰度信息仅基于其在输入中的原始灰度级别），或者将图像作为一个整体进行处理（整个图像）。

规格严格 功夫到家

空间域中的图像增强



像素

- 为每个像素提供**上下文信息**。
- **邻域内的关系**决定了图像特征。

规格严格 功夫到家



哈爾濱工業大學
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3. 圖像的濾波

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回顧

- 1975 年，福島邦彥，Cognitron（認知機）
☞ 一種早期的神經網絡模型，用於模式識別，被認為是卷積神經網絡的雛形。
- 1980 年，福島邦彥，Neocognitron（新認知機）
☞ 在認知機基礎上發展，引入分層結構，為後來的卷積神經網絡奠定基礎。
- 1989 年，Lecun，《泛化和網絡設計策略，反向傳播算法在手寫郵政編碼中的應用》☞ 將反向傳播算法應用於實際識別任務，是神經網絡應用的重要進展。
- 1998 年，Lecun，《基於梯度的學習在文檔識別中的應用》，LeNet-5
☞ 提出經典模型 LeNet-5，用於手寫字符識別，是現代卷積神經網絡的重要基礎。
- 2006 年，傑克·布夫里，《卷積神經網絡筆記》☞ 對卷積神經網絡進行系統性總結和介紹。
- 2012 年，Alex，《基於深度卷積神經網絡的 ImageNet 圖像分類》☞ 顯著提升圖像分類性能，推動深度學習發展。
- 2015 年，谷歌，《重新思考計算機視覺的 Inception 架構》，Inception V3
☞ 提出更高效的網絡結構，提高性能同時降低計算成本。
- 2015 年，《用於語義分割的反卷積網絡學習》，Deconv ☞ 用於像素級分類任務（語義分割）。
- 2018 年，Nvidia，《使用部分卷積修復不規則孔洞的圖像》☞ 用於圖像缺失區域的修復。
- Conformer、ConvNet、.....

規格嚴格 功夫到家

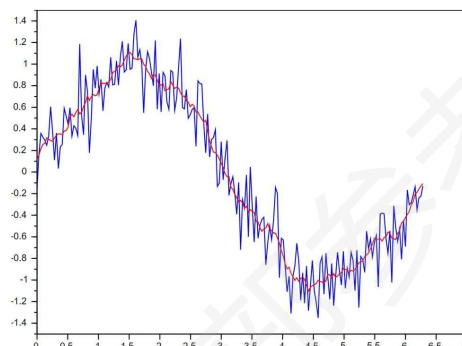
回顾：移动平均

$$I \in \mathcal{R}^{m \times 1}$$

$$h[n] = \frac{1}{k} \sum_{i=n-k+1}^n (1) I[i]$$

窗口大小 'k'

$$h[n] = \frac{1}{k} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}^T I[n-k+1:n]$$

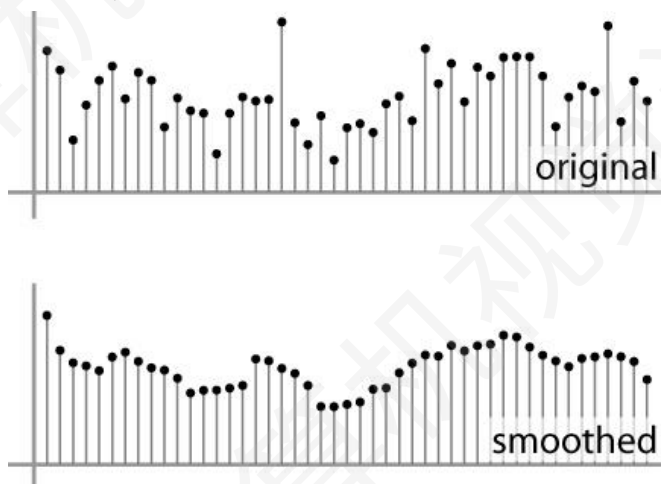


$$I[n - k/2 : n + k/2] \quad \text{不同窗口}$$

规格严格 功夫到家

回顾：一维信号的移动平均 (MA)

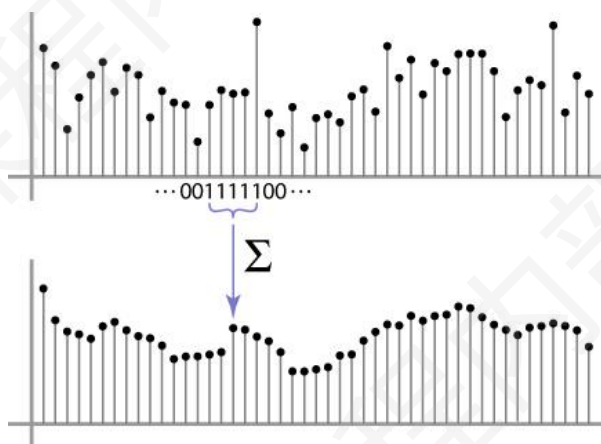
- 基本思路：通过对滑动窗口取平均值来定义一个新函数。
- 一个简单的例子：平滑处理



规格严格 功夫到家

回顾：一维信号的移动平均 (MA)

- 可以给移动平均线添加权重
 - 权重 $[1, 1, 1, 1, 1] / 5$
 - 信号 (输入) 与向量 (一维滤波器) 的点积



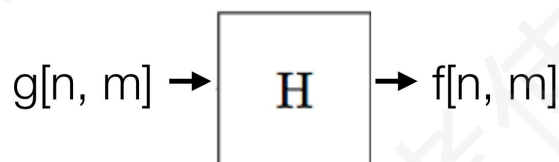
平滑处理

规格严格 功夫到家

3.1 二维滤波

规格严格 功夫到家

滤波



➤ 目标:

- 去除不必要的变化来源,
- 并保留与我们需要解决的任务相关的信息。

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什么是滤波？为什么要滤波？

➤ 滤波

- 生成一张像素值为原始像素值加权或组合而成的新图像

➤ 为什么要滤波？

- 从图片中获取有用信息
 - 例如，提取边缘或轮廓（以理解形状）

➤ 为了增强图像

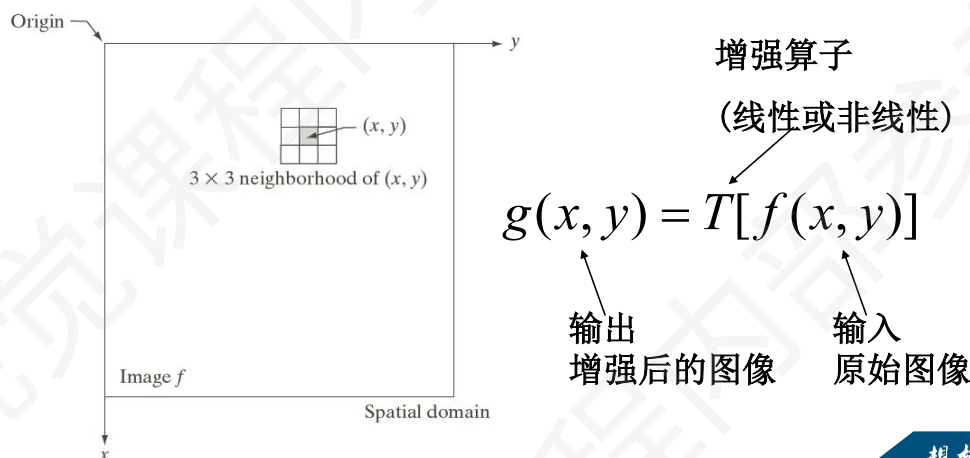
- 例如，去除噪声
- 例如，用于锐化或增强图像

➤ 卷积神经网络中的一个关键特征提取算子

规格严格 功夫到家

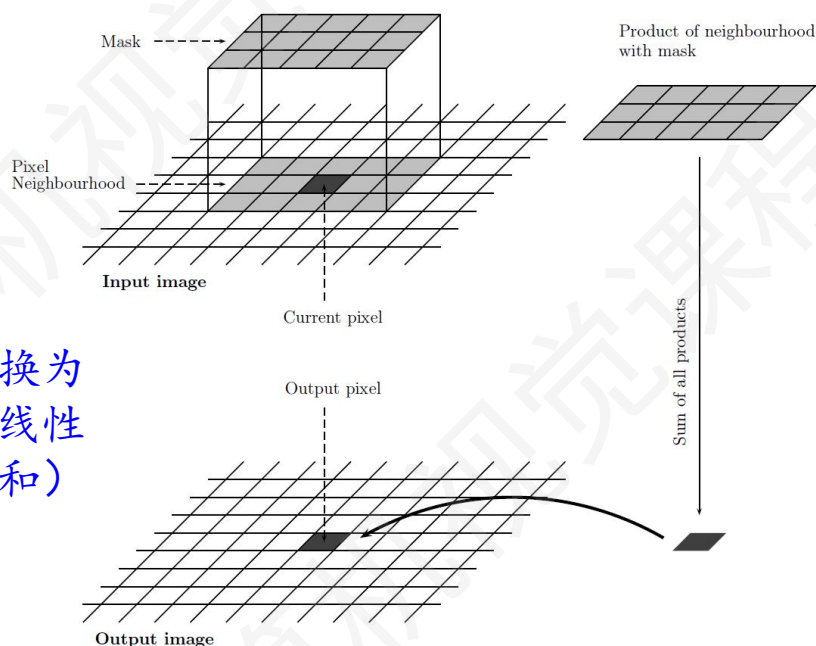
空间域图像增强

- 图像增强是直接基于图像的空间信息（包括像素位置和灰度）进行的。
- 增强算子是在像素邻域内定义的。



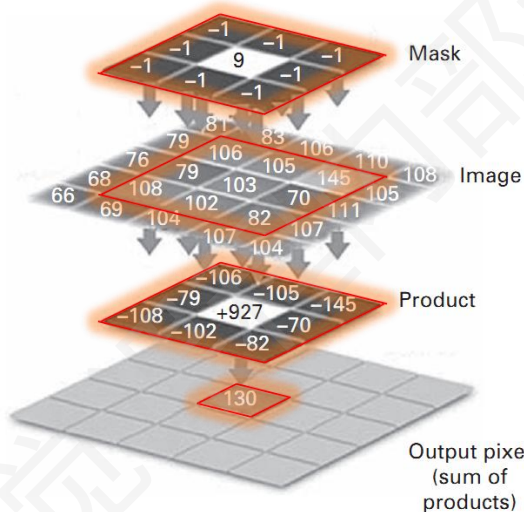
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将每个像素替换为
其邻域像素的线性
组合（即加权和）



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内核权重相加必须等于 1



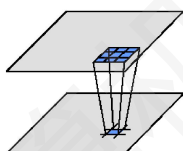
-1	-1	-1
-1	9	-1
-1	-1	-1

$$\begin{aligned}
 &(-1) \times 106 + (-1) \times 105 + (-1) \times 145 \\
 &+ (-1) \times 79 + (+9) \times 103 + (-1) \times 70 \\
 &+ (-1) \times 108 + (-1) \times 102 + (-1) \times 82 \\
 &= 130
 \end{aligned}$$

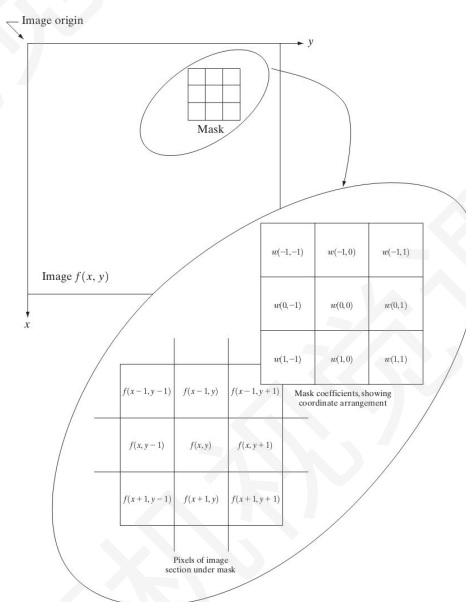
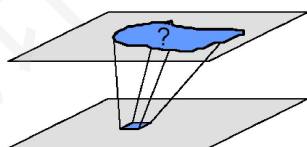
Digital convolution: the mask is placed over the source image, mask elements are multiplied by pixel values, and the results added to form an output pixel.

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规则映射



非规则映射



- ✓ Filter (滤波器)
- ✓ Mask (掩模)
- ✓ Kernel (核)
- ✓ Template (模板)
- ✓ Window (窗口)

规格严格 功夫到家

3.1 空间域中的图像滤波

➤ 空间滤波和空间滤波器的定义

- 使用空间模板进行图像处理称为空间滤波。
 - 滤波器是对数字网格进行数学运算
 - 模板本身称为空间滤波器

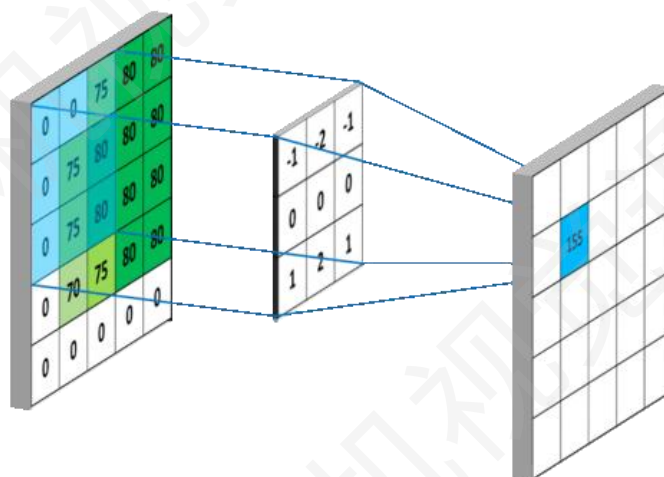
➤ 效果

- 平滑
- 锐化
- 测量纹理

- ✓ Filter (滤波器)
- ✓ Mask (掩模)
- ✓ Kernel (核)
- ✓ Template (模板)
- ✓ Window (窗口)

规格严格 功夫到家

3.1 空间域中的图像滤波



规格严格 功夫到家

盒式濾波器

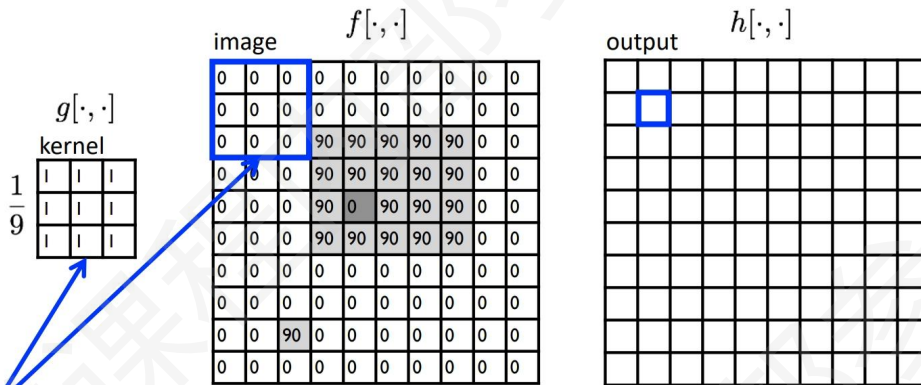
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示例：盒式濾波器

- 又称二维矩形濾波器
 - 又称平方均值濾波器
 - 用局部平均值替換像素
 - 具有平滑（模糊）效果
- $g[\cdot, \cdot]$

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$


規格嚴格 功夫到家

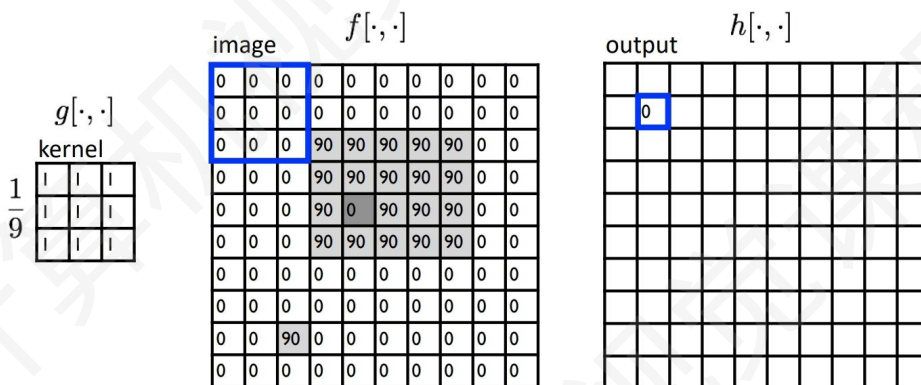


note that we assume that
the kernel coordinates
are centered

$$h[m, n] = \sum_{k, l} g[k, l] f[m + k, n + l]$$

output
 k, l
filter
image (signal)

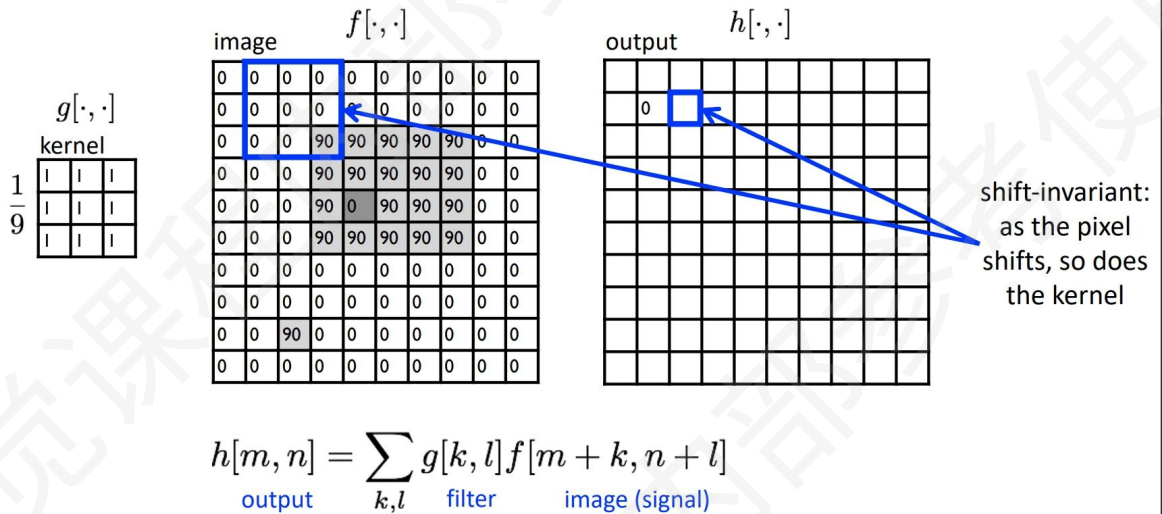
規格嚴格 功夫到家



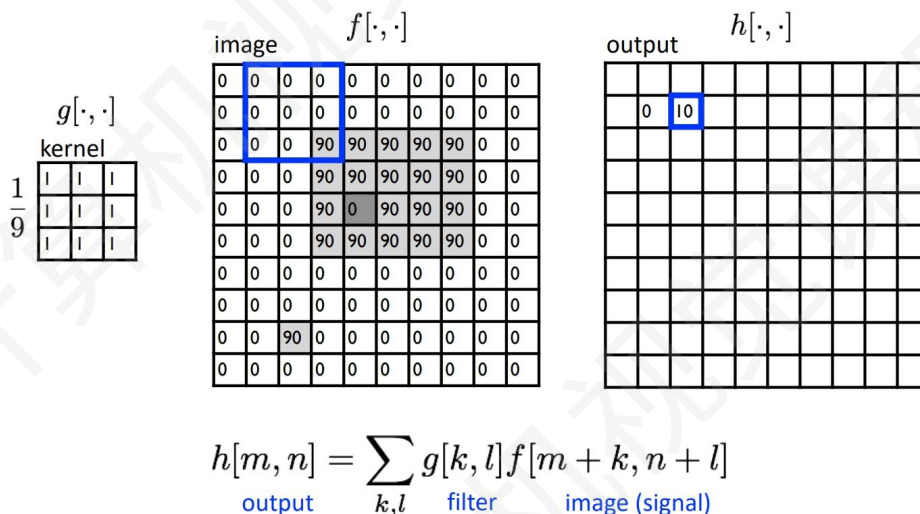
$$h[m, n] = \sum_{k, l} g[k, l] f[m + k, n + l]$$

output
 k, l
filter
image (signal)

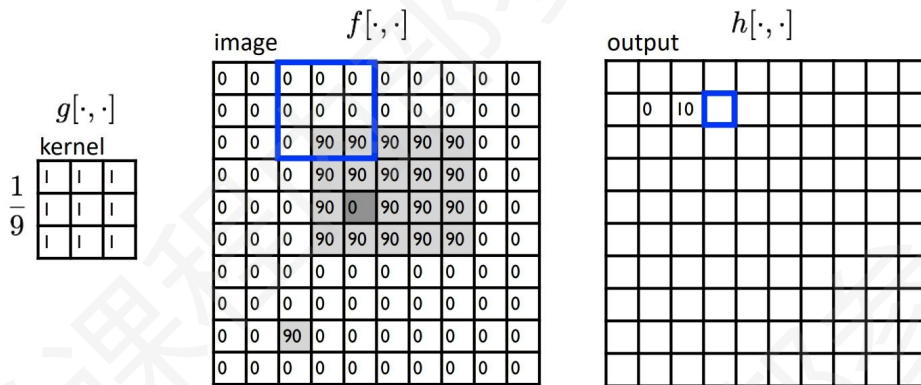
規格嚴格 功夫到家



規格嚴格 功夫到家



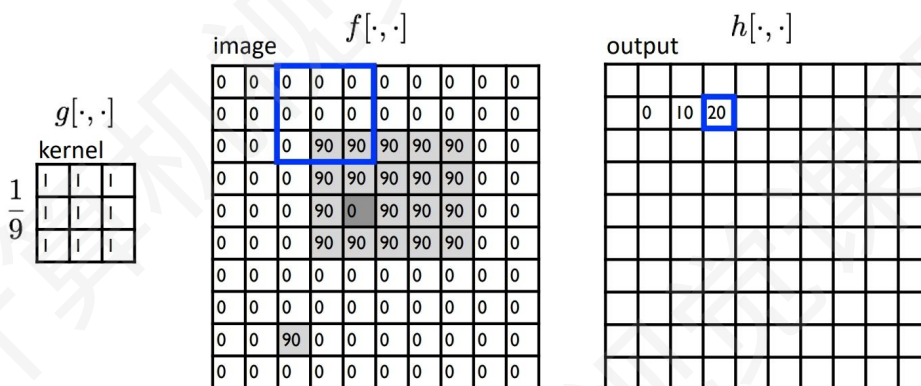
規格嚴格 功夫到家



$$h[m, n] = \sum_{k, l} g[k, l] f[m + k, n + l]$$

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 k, l
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image (signal)

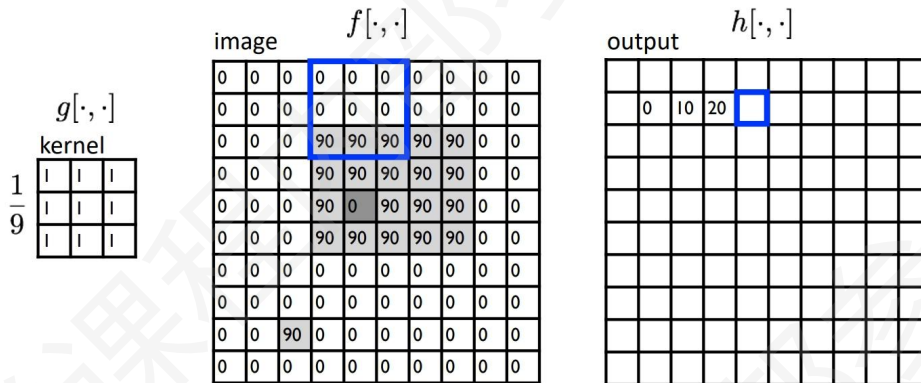
規格嚴格 功夫到家



$$h[m, n] = \sum_{k, l} g[k, l] f[m + k, n + l]$$

output
 k, l
filter
image (signal)

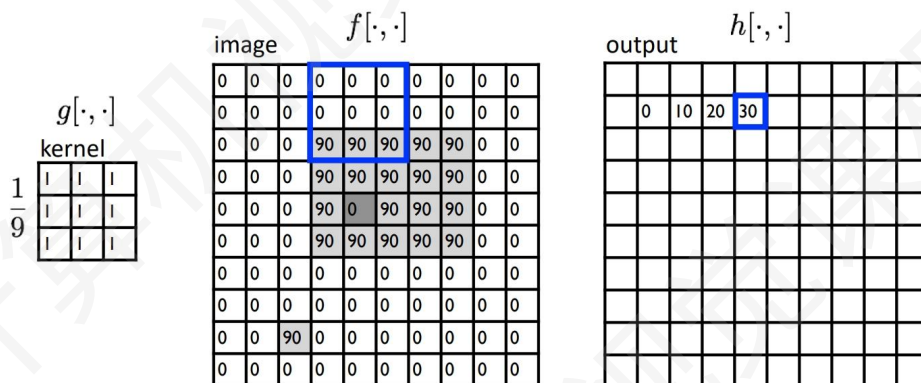
規格嚴格 功夫到家



$$h[m, n] = \sum_{k, l} g[k, l] f[m + k, n + l]$$

output
 k, l
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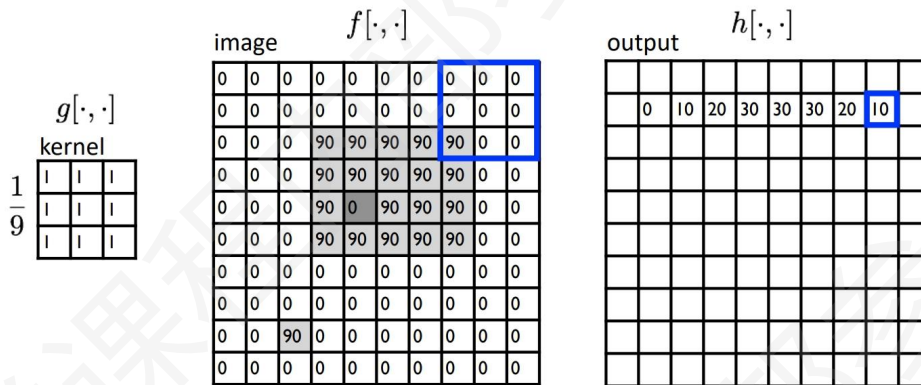
規格嚴格 功夫到家



$$h[m, n] = \sum_{k, l} g[k, l] f[m + k, n + l]$$

output
 k, l
filter
image (signal)

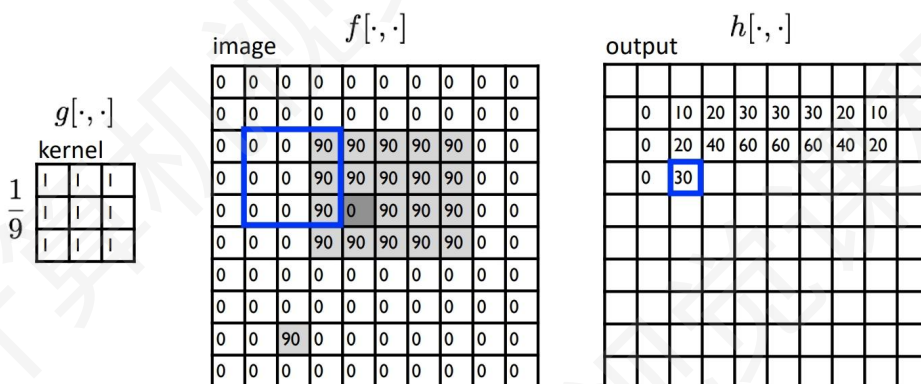
規格嚴格 功夫到家



$$h[m, n] = \sum_{k, l} g[k, l] f[m + k, n + l]$$

output
 k, l
filter
image (signal)

規格嚴格 功夫到家



$$h[m, n] = \sum_{k, l} g[k, l] f[m + k, n + l]$$

output
 k, l
filter
image (signal)

規格嚴格 功夫到家

		image $f[\cdot, \cdot]$											output $h[\cdot, \cdot]$											
$g[\cdot, \cdot]$ kernel $\frac{1}{9}$		0	0	0	0	0	0	0	0	0	0													
		0	0	0	0	0	0	0	0	0	0		0	10	20	30	30	30	20	10				
		0	0	0	90	90	90	90	90	0	0		0	20	40	60	60	60	40	20				
		0	0	0	90	90	90	90	90	0	0		0	30	50	80	80	90	60	30				
		0	0	0	90	0	90	90	90	0	0		0	30	50	80	80	90	60	30				
		0	0	0	90	90	90	90	90	0	0		0	20	30	50	50	60	40	20				
		0	0	0	0	0	0	0	0	0	0		0	10	20	30	30	30	20	10				
		0	0	0	0	0	0	0	0	0	0		10	10	10	10	0	0	0	0				
		0	0	90	0	0	0	0	0	0	0		10											
		0	0	0	0	0	0	0	0	0	0													

$$h[m, n] = \sum_{k, l} g[k, l] f[m + k, n + l]$$

output
 k, l
filter
image (signal)

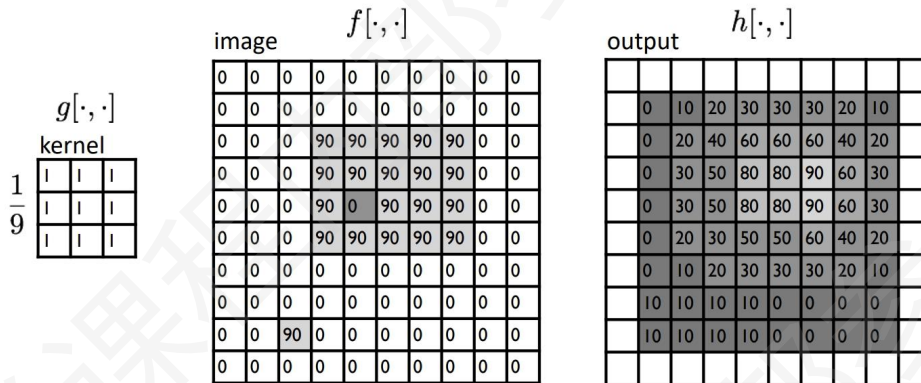
規格嚴格 功夫到家

		image $f[\cdot, \cdot]$											output $h[\cdot, \cdot]$											
$g[\cdot, \cdot]$ kernel $\frac{1}{9}$		0	0	0	0	0	0	0	0	0	0													
		0	0	0	0	0	0	0	0	0	0		0	10	20	30	30	30	20	10				
		0	0	0	90	90	90	90	90	0	0		0	20	40	60	60	60	40	20				
		0	0	0	90	90	90	90	90	0	0		0	30	50	80	80	90	60	30				
		0	0	0	90	0	90	90	90	0	0		0	30	50	80	80	90	60	30				
		0	0	0	90	90	90	90	90	0	0		0	20	30	50	50	60	40	20				
		0	0	0	0	0	0	0	0	0	0		0	10	20	30	30	30	20	10				
		0	0	0	0	0	0	0	0	0	0		10	10	10	10	0	0	0	0				
		0	0	90	0	0	0	0	0	0	0		10	10	10	10	0	0	0	0				
		0	0	0	0	0	0	0	0	0	0													

$$h[m, n] = \sum_{k, l} g[k, l] f[m + k, n + l]$$

output
 k, l
filter
image (signal)

規格嚴格 功夫到家



$$h[m, n] = \sum_{k, l} g[k, l] f[m + k, n + l]$$

output
 k, l
filter
image (signal)

規格嚴格 功夫到家

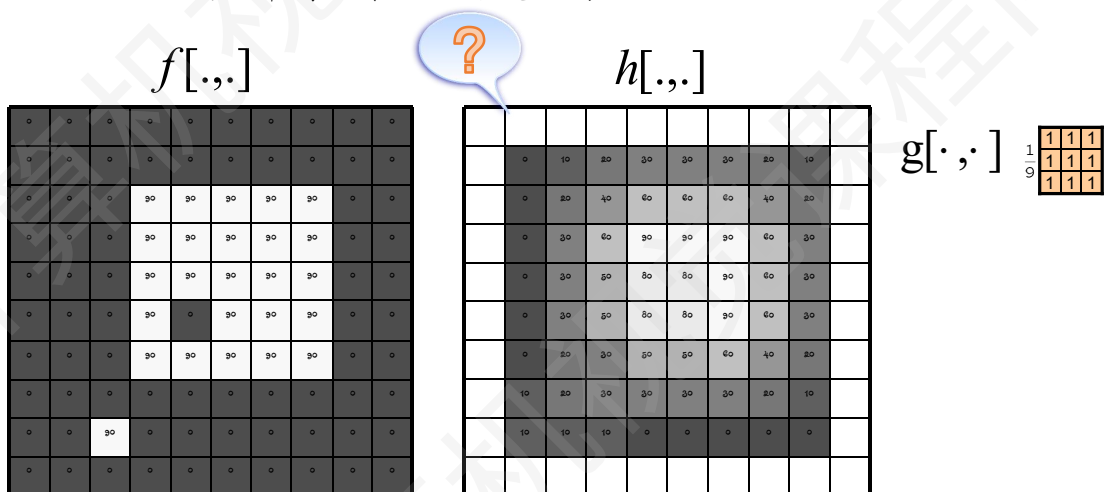
3.1 滤波

3.2 卷积

3.3 边缘检测

它有什么作用？

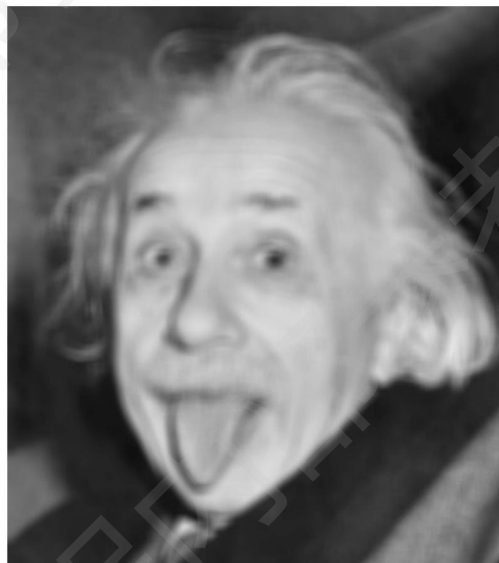
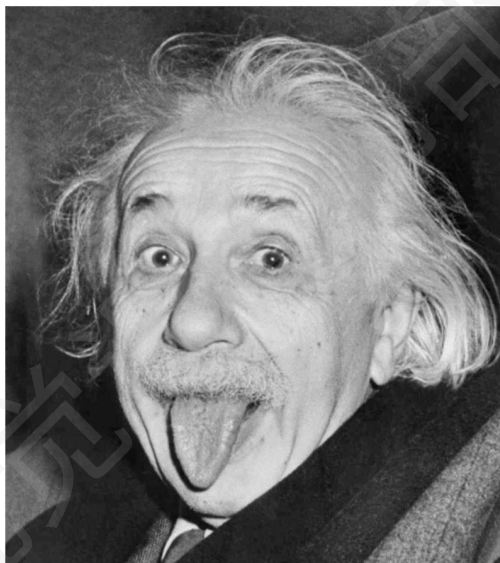
- 将每个像素替换为其邻域像素的平均值
- 实现平滑效果（去除尖锐特征）



$$h[m, n] = \sum_{k, l} g[k, l] f[m + k, n + l]$$

規格嚴格 功夫到家

使用盒式滤波器进行平滑处理



規格嚴格 功夫到家

使用盒式滤波器进行平滑处理



規格嚴格 功夫到家

3.2 二维卷积

规格严格 功夫到家

滤波 vs. 卷积

➤ 二维滤波

f =滤波器, 维度 $k \times l$

I =图像, 维度 $m \times n$

$$h[m,n] = \sum_{k,l} f[k,l] I[m+k,n+l]$$

➤ 二维卷积

$$h[m,n] = \sum_{k,l} f[k,l] I[m-k,n-l]$$

规格严格 功夫到家

二维离散信号的卷积

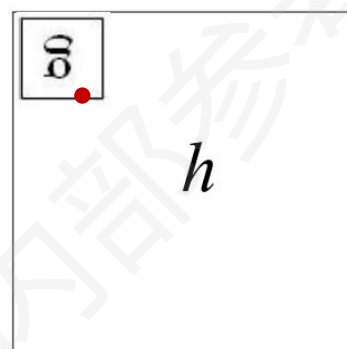
Definition of filtering as convolution:

$$(f * I)(x, y) = \sum_{i, j=-\infty}^{\infty} f(i, j) I(x - i, y - j)$$

filtered image $\rightarrow$ $(f * I)(x, y)$ $\leftarrow$ filter $\leftarrow$ input image I
 notice the flip



Convention:
kernel is “flipped”



卷积的核会被**翻转**:

- 满足很多重要数学性质 (交换律、结合律)
- 是经典的线性时不变系统 (LTI) 响应形式

规格严格 功夫到家

互相关

设 F 为输入图像, H 为核 (大小为 $2k+1 \times 2k+1$),
 G 为输出图像:

$$G[i, j] = \sum_{u=-k}^k \sum_{v=-k}^k H[u, v] F[i + u, j + v]$$

这称为**互相关运算**:

$$G = H \otimes F$$

- 可视为每个像素处局部邻域与卷积核的“点积”

规格严格 功夫到家

卷积 vs. 互相关

➤ 滤波作为卷积的定义：

注意翻转

$$(f * I)(x, y) = \sum_{i, j=-\infty}^{\infty} f(i, j) I(x - i, y - j)$$

➤ 滤波作为互相关的定义：

注意缺少翻转

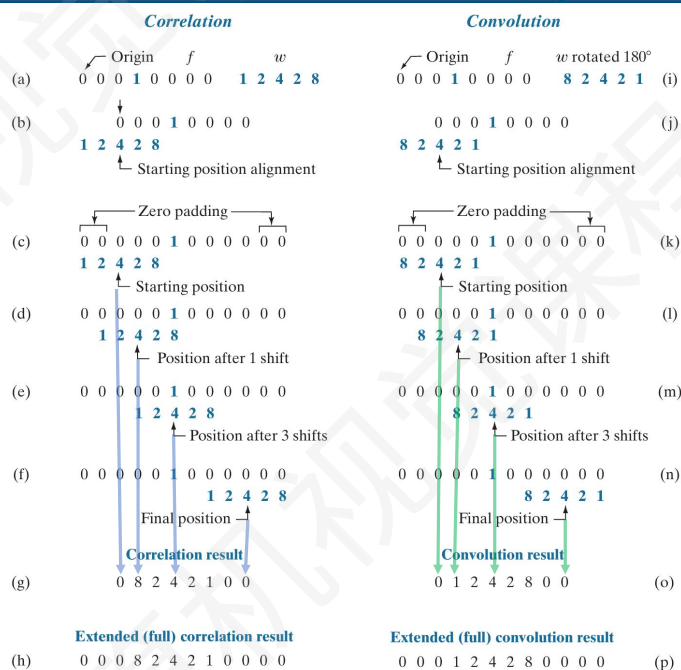
$$(f * I)(x, y) = \sum_{i, j=-\infty}^{\infty} f(i, j) I(x + i, y + j)$$

大多数时候无关紧要，因为我们的核函数是对称的。

规格严格 功夫到家

卷积 vs. 互相关

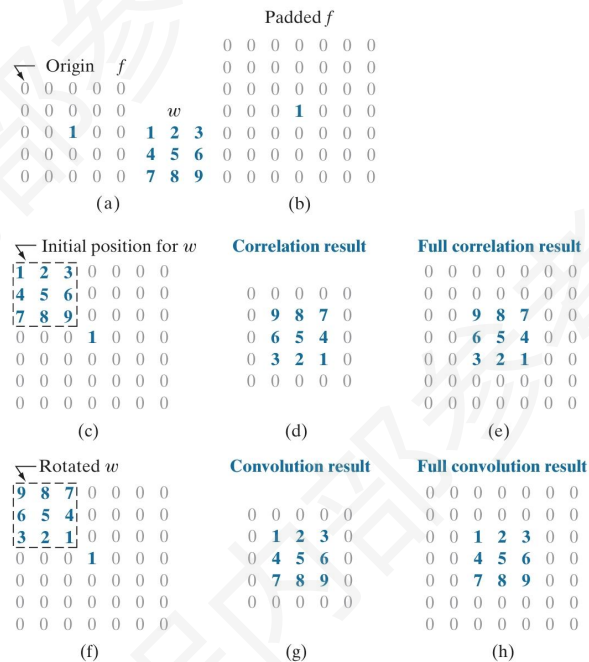
—— 从一维视角看



规格严格 功夫到家

卷积 vs. 互相关

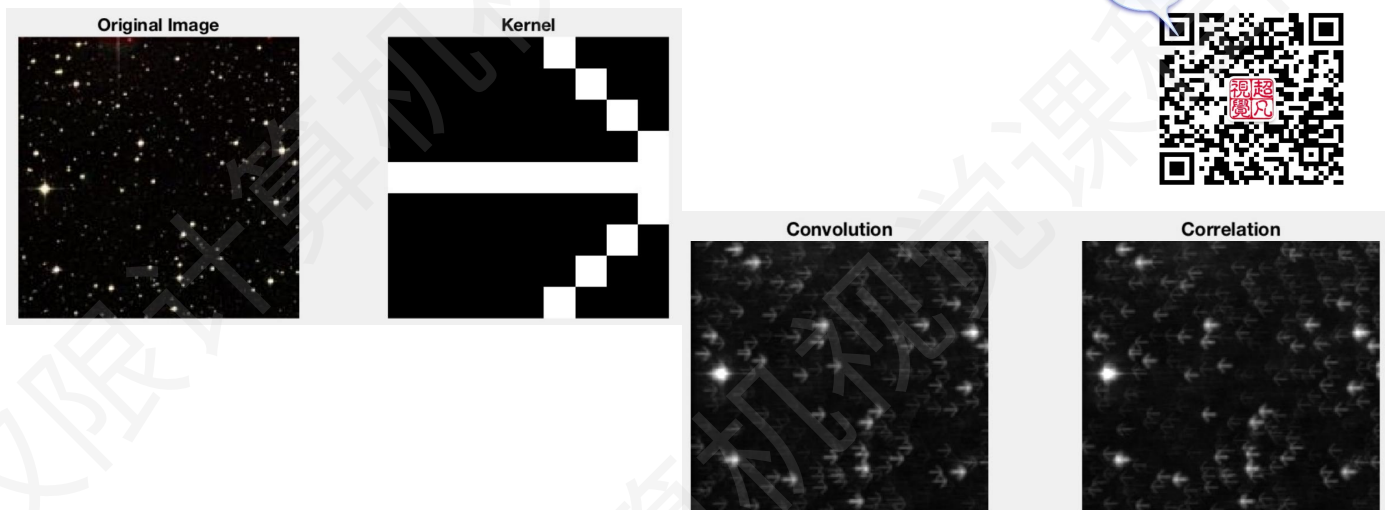
— 从二维视角看



规格严格 功夫到家

卷积 vs 互相关 — 关于翻转

Note that the direction of the arrow flips when going from convolution to correlation.



规格严格 功夫到家

3.3 Conv 作为线性滤波器的关键性质

- 平移不变性
- 可分离滤波器

規格嚴格 功夫到家

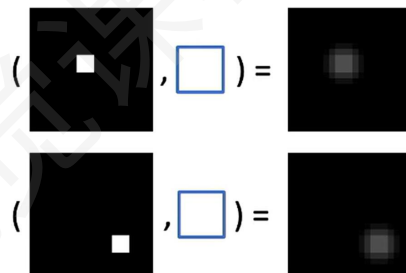
线性滤波器的关键性质

□ 线性:

$$\text{imfilter}(I, f_1 + f_2) = \text{imfilter}(I, f_1) + \text{imfilter}(I, f_2)$$

□ 平移不变性: 无论像素位置如何, 行为一致

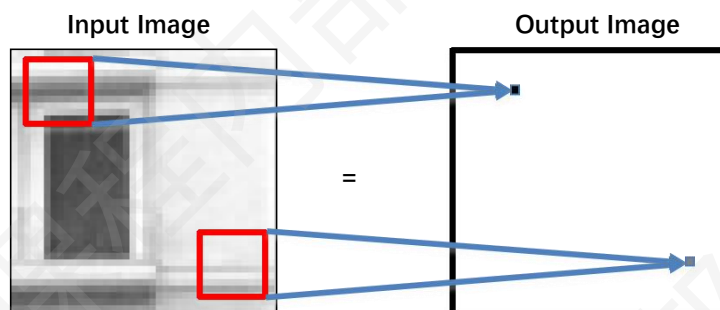
$$\text{imfilter}(I, \text{shift}(f)) = \text{shift}(\text{imfilter}(I, f))$$



□ 任何线性、平移不变的算子都可以表示为卷积

規格嚴格 功夫到家

A translation invariant filter



The same weighting occurs within each window

規格嚴格 功夫到家

可分离滤波器

- 卷积作为线性滤波器

規格嚴格 功夫到家

可分离滤波器

Conv(vector, transposed vector) → **outer product**

$$\begin{array}{|c|} \hline \text{Fy1} \\ \hline \text{Fy2} \\ \hline \text{Fy3} \\ \hline \end{array}
 * \begin{array}{|c|c|c|} \hline \text{Fx1} & \text{Fx2} & \text{Fx3} \\ \hline \end{array}
 = \begin{array}{|c|c|c|} \hline \text{Fx1} * \text{Fy1} & \text{Fx2} * \text{Fy1} & \text{Fx3} * \text{Fy1} \\ \hline \text{Fx1} * \text{Fy2} & \text{Fx2} * \text{Fy2} & \text{Fx3} * \text{Fy2} \\ \hline \text{Fx1} * \text{Fy3} & \text{Fx2} * \text{Fy3} & \text{Fx3} * \text{Fy3} \\ \hline \end{array}$$

(Using “full” convolution with zero padding)
(Also ignoring filter flips)

规格严格 功夫到家

可分离滤波器

- 如果一个二维滤波器可以表示为一个“列”与一个“行”的乘积，则它是可分离的。

example:
box filter

$$\begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}
 = \begin{array}{|c|} \hline 1 \\ \hline 1 \\ \hline 1 \\ \hline \end{array}
 * \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline \end{array}$$

column row

- 该滤波矩阵的秩是多少？
 - 为何重要？
 - 使用可分离滤波器的二维卷积等价于**2次一维卷积**（分别使用“列”滤波器和“行”滤波器）

规格严格 功夫到家

Recap: Separability Example

2D convolution
(center location only)

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 3 & 3 \\ 3 & 5 & 5 \\ 4 & 4 & 6 \end{bmatrix}$$

The filter factors
into a product of 1D
filters:

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

Perform convolution
along rows:

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 3 & 3 \\ 3 & 5 & 5 \\ 4 & 4 & 6 \end{bmatrix} = \begin{bmatrix} & 11 & \\ & 18 & \\ & 18 & \end{bmatrix}$$

Followed by convolution
along the remaining column:

Source: K. Grauman

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可分离滤波器

- 如果一个二维滤波器可以表示为一个“列”与一个“行”的乘积，则它是可分离的。

example:
box filter

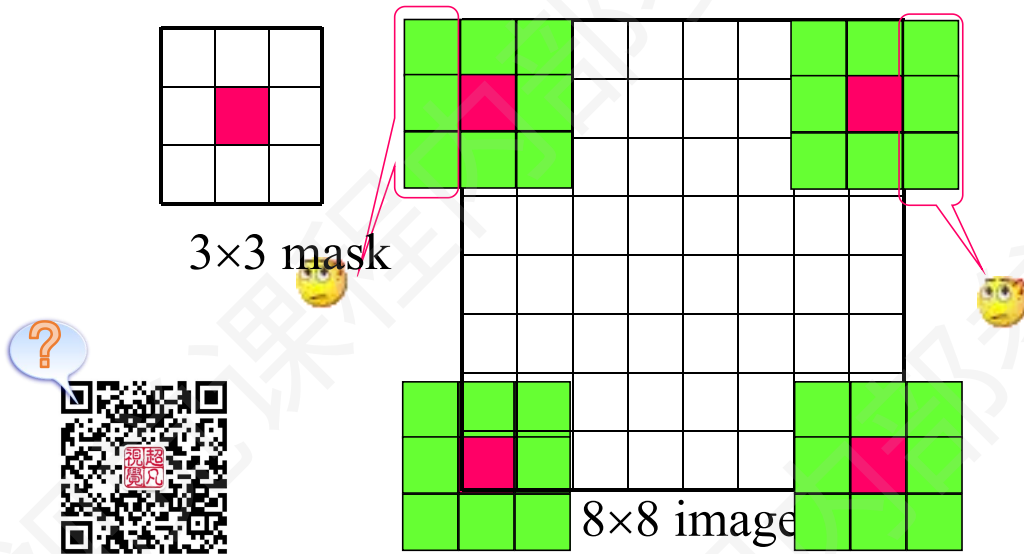
$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

column row

- 如果图像有 $M \times M$ 个像素，滤波器内核的大小为 $N \times N$ ：
- 使用不可分离滤波器进行卷积的代价是什么？ $\rightarrow M^2 \times N^2$
- 使用可分离滤波器进行卷积的代价是什么？ $\rightarrow M^2 \times 2 \times N$

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Handling boundaries ??



規格嚴格 功夫到家

滤波中的图像边界问题：Padding

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Input

i11	i12	i13	i14	i15	i16
i21	i22	i23	i24	i25	i26
i31	i32	i33	i34	i35	i36
i41	i42	i43	i44	i45	i46
i51	i52	i53	i54	i55	i56

Filter

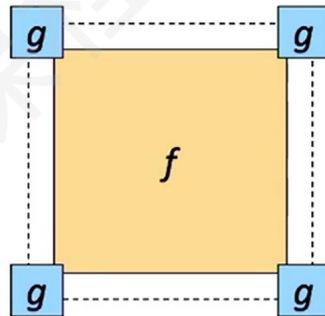
f11	f12	f13
f21	f22	f23
f31	f32	f33

Output

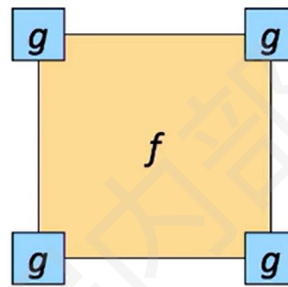
o11	o12	o13	o14
o21	o22	o23	o24
o31	o32	o33	o34

Convolution doesn't keep the whole image.
 Suppose f is the image and g the filter.

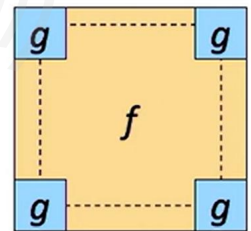
Full: Any part
 of g touches f .



Same: Output
 is same size as f

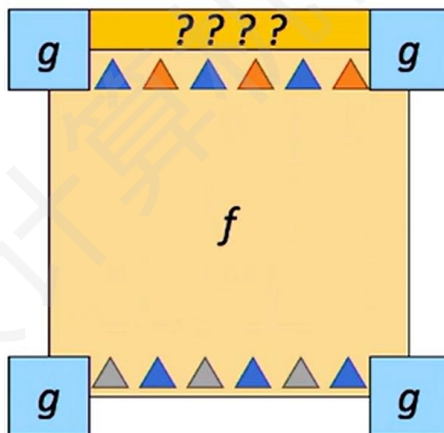


Valid: Filter doesn't
 fall off edge



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What to about the “?” region?



Symm: fold sides over



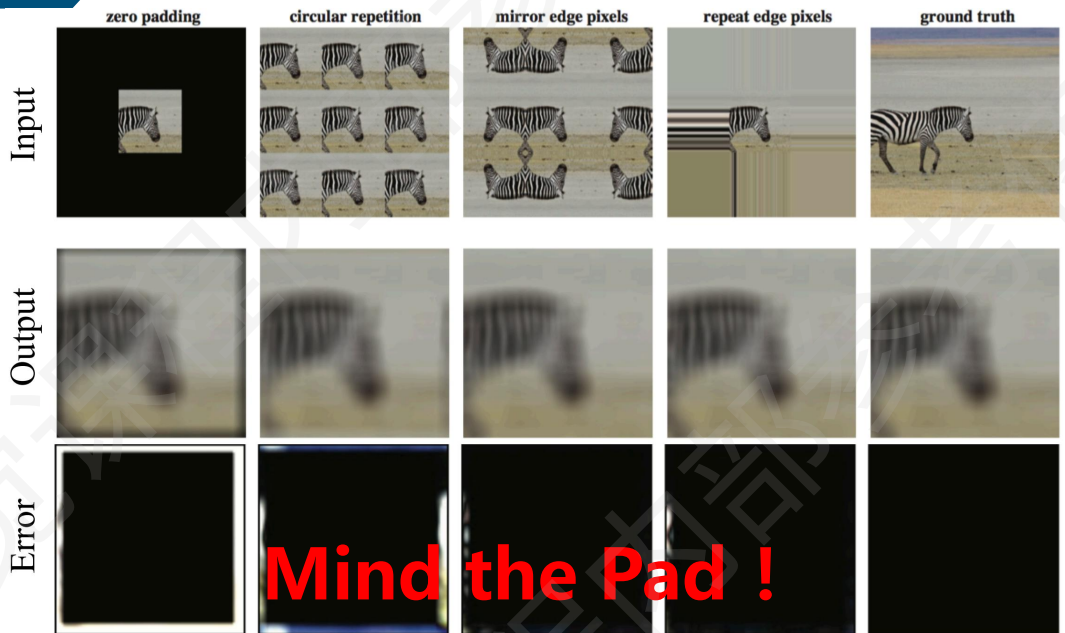
Circular/Wrap: wrap around



pad/fill: add value, often 0



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Mind the Pad -- CNNs Can Develop Blind Spots



Bilal Alsallakh, Narine Kokhlikyan, Vivek Miglani, Jun Yuan, Orion Reblitz-Richardson

 Published: 13 Jan 2021, Last Modified: 12 Oct 2025 ICLR 2021 Spotlight Readers:  Everyone Show Bibtex Show Revisions

Keywords: CNN, convolution, spatial bias, blind spots, foveation, padding, exposition, debugging, visualization

Abstract: We show how feature maps in convolutional networks are susceptible to spatial bias. Due to a combination of architectural choices, the activation at certain locations is systematically elevated or weakened. The major source of this bias is the padding mechanism. Depending on several aspects of convolution arithmetic, this mechanism can apply the padding unevenly, leading to asymmetries in the learned weights. We demonstrate how such bias can be detrimental to certain tasks such as small object detection: the activation is suppressed if the stimulus lies in the impacted area, leading to blind spots and misdetection. We explore alternative padding methods and propose solutions for analyzing and mitigating spatial bias.

One-sentence Summary: The padding mechanism in CNNs can induce harmful spatial bias in the learned weights and in the feature maps, which can be mitigated with careful architectural choices.

Supplementary Material:  zip

Code Of Ethics: I acknowledge that I and all co-authors of this work have read and commit to adhering to the ICLR Code of Ethics

Community Implementations:  2 code implementations

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空间滤波器

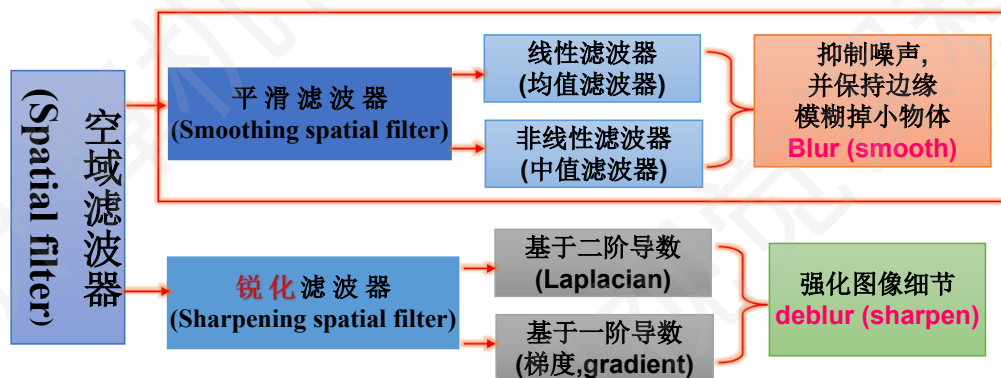
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3.1 滤波

3.2 卷积

3.3 边缘检测

空间滤波器



規格嚴格 功夫到家

平滑滤波器

	a				b		
	1	1	1		1	2	1
$\frac{1}{9} \times$	1	1	1		2	4	2
	1	1	1	$\frac{1}{16} \times$	1	2	1

为什么该滤波器能去除（抑制）噪声？

实际图像=理想图像+噪声

$$g(x, y) = f(x, y) + \eta(x, y)$$

在理想图像中，邻域内的像素是相似的，但邻域中的噪声是随机的

通过平均多幅图像也可实现图像去噪

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空间滤波器

➤ 平滑滤波器：

- 模糊处理：从图像中移除无关节节。
- 降噪：降低噪音。

➤ 平滑空间滤波器的分类：

➤ 线性滤波器： $R = w_1 z_1 + w_2 z_2 + \dots + w_{mn} z_{mn} = \sum_{i=1}^{mn} w_i z_i$

➤ 均值滤波器、加权均值滤波器（如高斯滤波器）

➤ 非线性滤波器： $R = w_1 z_1 + w_2 z_2 + \dots + w_{mn} z_{mn} \neq \sum_{i=1}^{mn} w_i z_i$

➤ 中值滤波器、取八邻域平均值、取中值滤波器等（属于排序统计滤波器）

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空间滤波器

➤ 中值滤波器

➤ 主要用途：去除**噪声**

➤ 计算公式： $R = \text{mid} \{z_k \mid k = 1, 2, \dots, n\}$

➤ 最大值滤波器

➤ 主要用途：寻找**最亮点**

➤ 计算公式： $R = \max \{z_k \mid k = 1, 2, \dots, n\}$

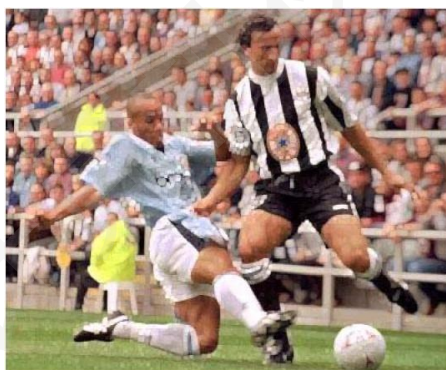
➤ 最小值滤波器

➤ 主要用途：寻找**最暗点**

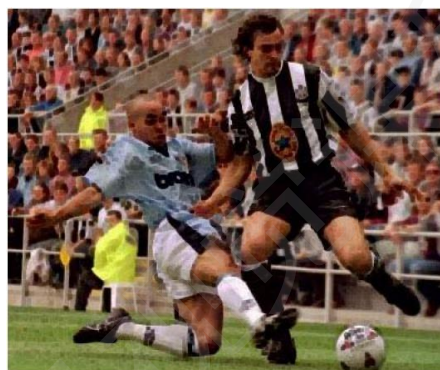
➤ 计算公式： $R = \min \{z_k \mid k = 1, 2, \dots, n\}$

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非线性滤波器



Match: City v. Newcastle United, 1992. Newcastle's Richard Kight tackled Manchester City's David Ginola during the match. City lost the match 3-1. Picture: REUTERS



Man City's Richard Kight tackles Newcastle's David Ginola at St James Park. City lost the match 3-1. Picture: REUTERS

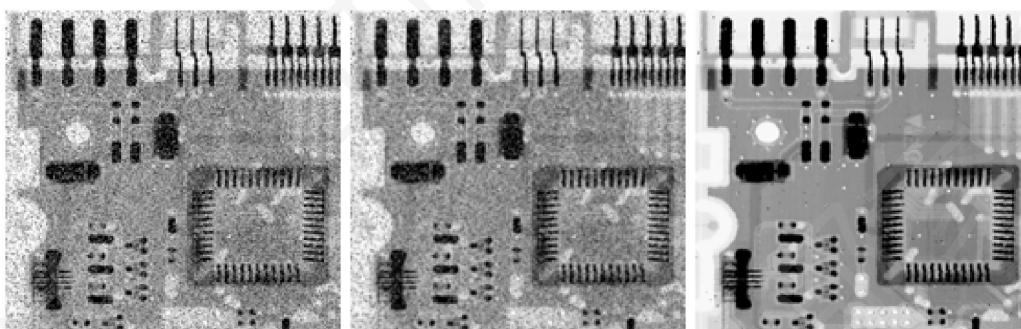
最大值滤波器

最小值滤波器

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中值滤波

椒盐噪声 (Salt-and-Pepper Noise) 是图像中随机出现的极端像素值——通常是纯黑 (0) 和纯白 (255) 的点, 就像撒在图像上的“盐粒和胡椒粒”。



(a) 受椒盐噪声污染的电路板 X 射线图像。

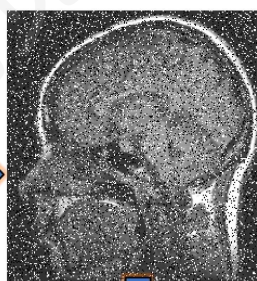
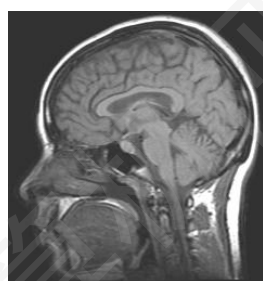
(b) 使用 3×3 均值滤波器进行降噪。

(c) 使用 3×3 中值滤波器进行降噪。

✓ 中值滤波器比均值滤波器更适合于椒盐噪声

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中值滤波

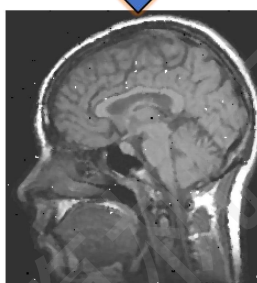


施加概率为0.2的椒盐噪声后的图像

噪声情况?

边缘情况?

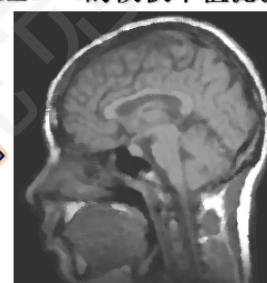
经 5×5 的模板中值滤波后



为什么还会出现
亮点和暗点?

怎么改进?

经 3×3 的模板中值滤波后



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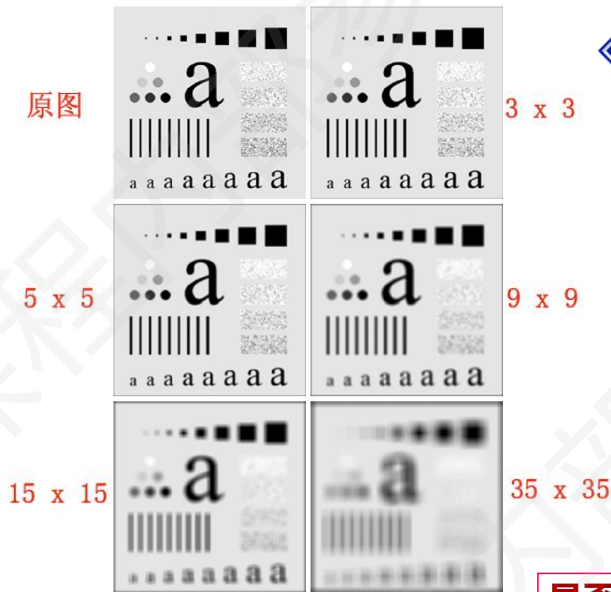


平滑滤波器

抑制噪声,

但问题也很严重:

模糊边缘/细节(blending effect)



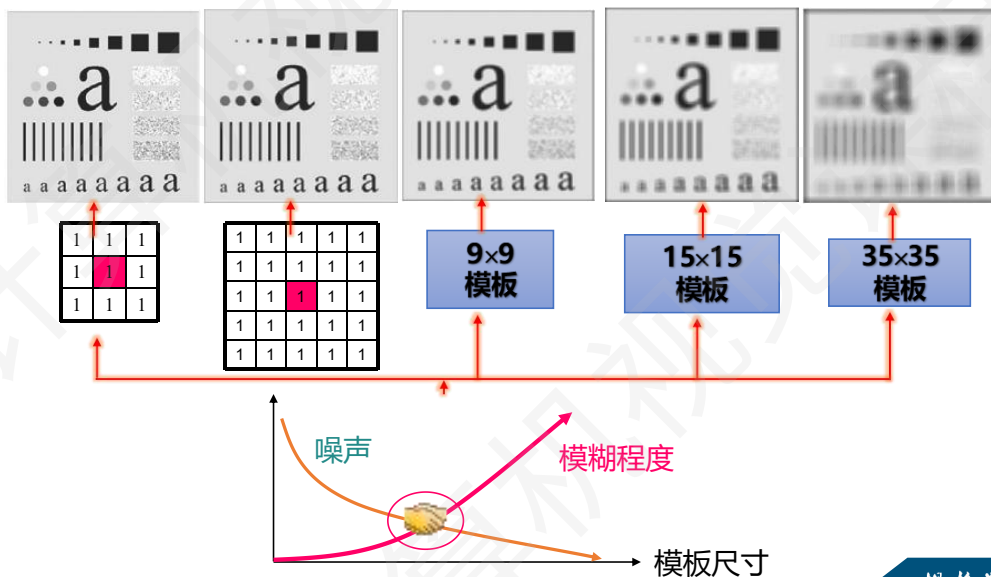
◆ 由于图像边缘是由图像灰度尖锐变化来表现的, 所以也存在边缘模糊的问题

是否存在问题?

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空间滤波器



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回顾: 盒式滤波器

- 也称为二维矩形滤波器
- 也称为平方均值滤波器
 - 用局部平均值替换像素
 - 具有平滑（模糊）效果

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

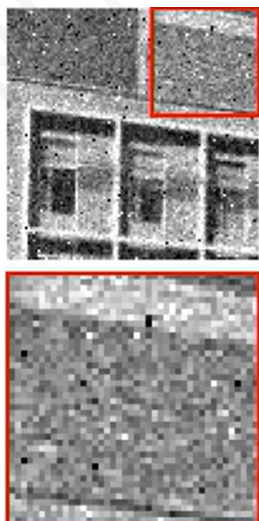


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盒式平滑

- 直觉：盒式滤波器对窗口内像素赋予相同权重，如果碰到噪声像素，它就会参与计算。

输入



盒式滤波器



为什么出现这种现象？
接下来如何改进？

问题：
块状
伪影

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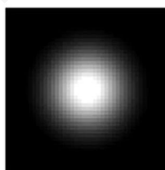
高斯模糊 vs 盒式模糊

盒式平滑
 $Filter_{ij} \propto 1$

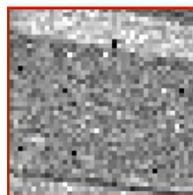
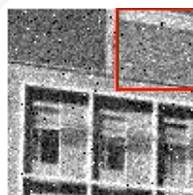


改进方法

$$Filter_{ij} \propto \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$



输入



盒式滤波器



高斯滤波器



➤ 直觉：根据与中心点的接近程度来赋予权重

还有一些斑点，但已经不是块状伪影了。

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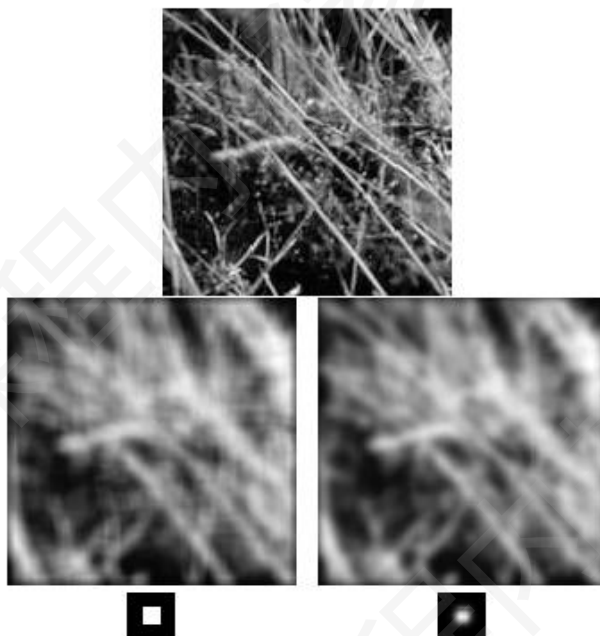
高斯滤波器

$$Filter_{ij} \propto \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$

$$\begin{bmatrix} 0.003 & 0.013 & 0.022 & 0.013 & 0.003 \\ 0.013 & 0.060 & 0.098 & 0.060 & 0.013 \\ 0.022 & 0.098 & 0.162 & 0.098 & 0.022 \\ 0.013 & 0.060 & 0.098 & 0.060 & 0.013 \\ 0.003 & 0.013 & 0.022 & 0.013 & 0.003 \end{bmatrix}$$



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規格嚴格 功夫到家

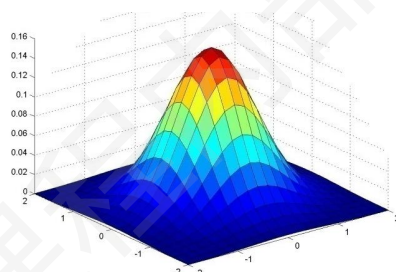
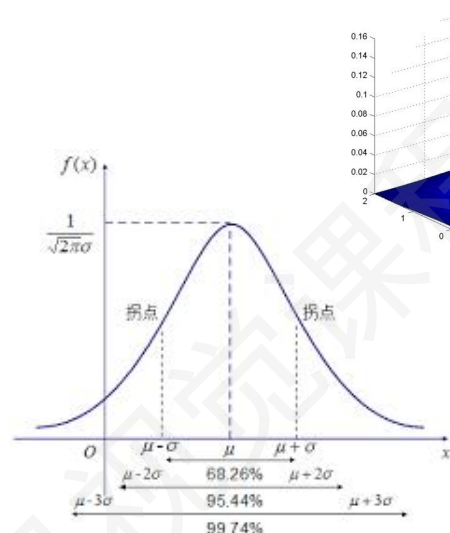
高斯核函数

- 最受欢迎的核函数之一

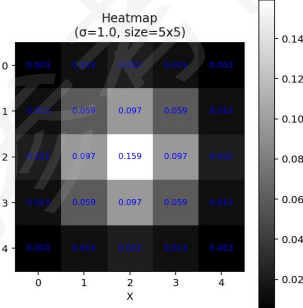
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高斯核函数

$$G_{\sigma} = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$



0.003 0.013 0.022 0.013 0.003
0.013 0.059 0.097 0.059 0.013
0.022 0.097 0.159 0.097 0.022
0.013 0.059 0.097 0.059 0.013
0.003 0.013 0.022 0.013 0.003



$\sigma=1$

滤波器尺寸为5x5

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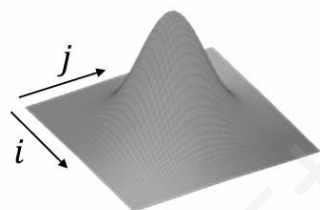
3.1 滤波

3.2 卷积

3.3 边缘检测

$\sigma = 8$, size = 21

$\sigma = 8$, size = 43



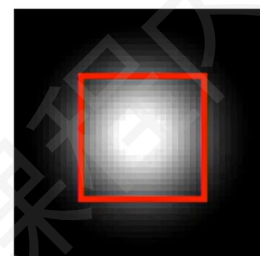
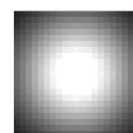
$$n_{\sigma}[i, j] = \frac{1}{2\pi\sigma^2} e^{-\frac{1}{2}\left(\frac{i^2+j^2}{\sigma^2}\right)}$$

$\sigma = 1$
filter = 21x21

$\sigma = 2$
filter = 21x21

$\sigma = 4$
filter = 21x21

$\sigma = 8$
filter = 21x21



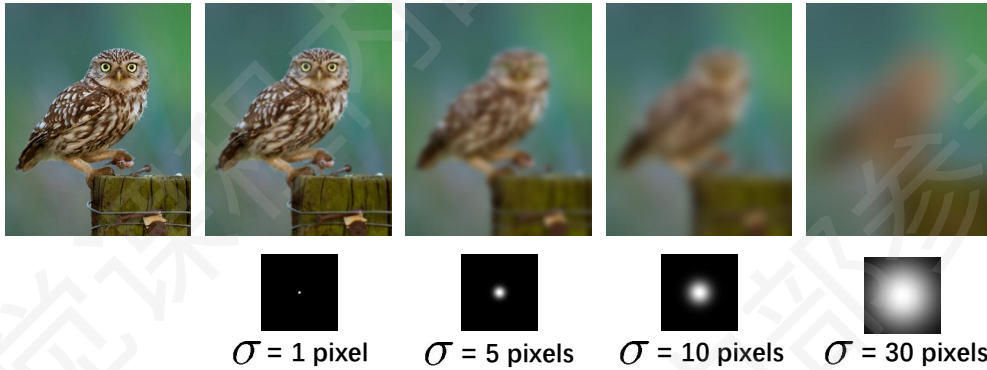
滤波器尺寸过小 — 近似效果差
期望尺寸 $\approx 6\sigma$ (99.7%的能量)
左边太小; 右边也略小!

推荐尺寸: $6\sigma + 1$

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高斯滤波器

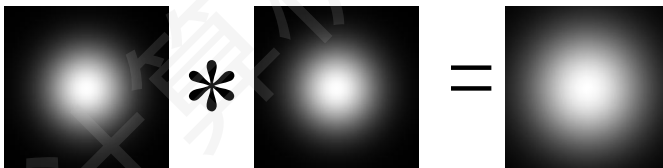


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Gaussian filter

- Removes “high-frequency” components from the image (low-pass filter)
- Convolution with self is another Gaussian



$$Filter_{ij} \propto \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$

→

$$Filter_{ij} \propto \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right) \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{y^2}{2\sigma^2}\right)$$

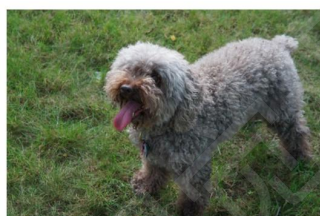
- Convolving twice with Gaussian kernel of width σ
convolving once with kernel of width $\sigma\sqrt{2}$

$$\sigma \sqrt{2} = \dots, \sigma' = \sqrt{n}\sigma$$

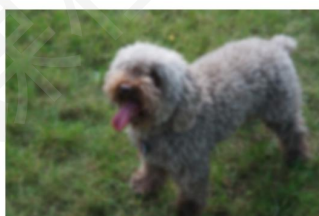
Source: K. Grauman 规格严格 功夫到家

➤ Convolution with self is another Gaussian

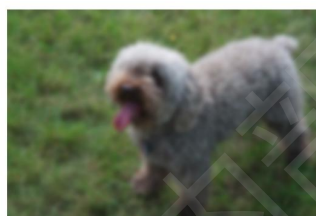
- Can smooth with small- σ kernel, **repeat**, get same result as larger- σ
- Convolving two times with Gaussian kernel with std. dev. σ is same as convolving once with kernel with std. dev. $\sqrt{2}\sigma$



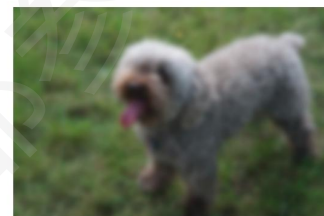
I



blur(blur(I))



blur(blur(blur(I)))



blur(blur(blur(blur(I))))

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可分离滤波器

使用一个二维高斯滤波器，等价于使用2个一维高斯滤波器

$$1D \text{ Gaussian} * 1D \text{ Gaussian} = 2D \text{ Gaussian}$$

$$\begin{aligned} \text{Image} * 2D \text{ Gauss} &= \text{Image} * (1D \text{ Gauss} * 1D \text{ Gauss}) \\ &= (\text{Image} * 1D \text{ Gauss}) * 1D \text{ Gauss} \end{aligned}$$

$$f * \begin{array}{c} \text{2D Gaussian Kernel} \\ \leftarrow K \rightarrow \end{array} = f * \begin{array}{c} \text{1D Gaussian Kernel} \\ \updownarrow K \end{array} * \begin{array}{c} \text{1D Gaussian Kernel} \\ \leftarrow K \rightarrow \end{array}$$

哪一个更快，为什么？

K^2 Multiplications

$K^2 - 1$ Additions

$2K$ Multiplications

$2(K - 1)$ Additions

規格嚴格 功夫到家

可分离滤波器

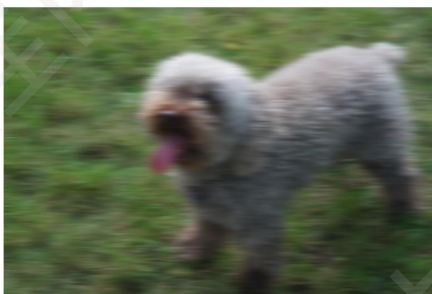
➤ 高斯滤波器是可分离核

➤ 先沿一个方向进行一维高斯模糊，然后再沿另一个方向进行。

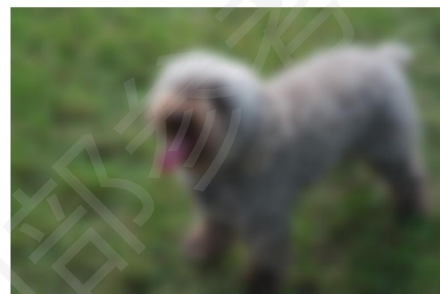
➤ 计算速度更快。对于一个 $n \times n$ 的核，计算复杂度是 $O(n)$ 而不是 $O(n^2)$



I

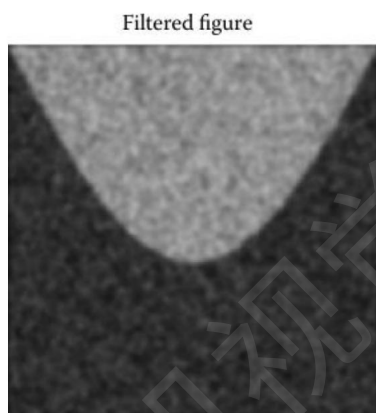
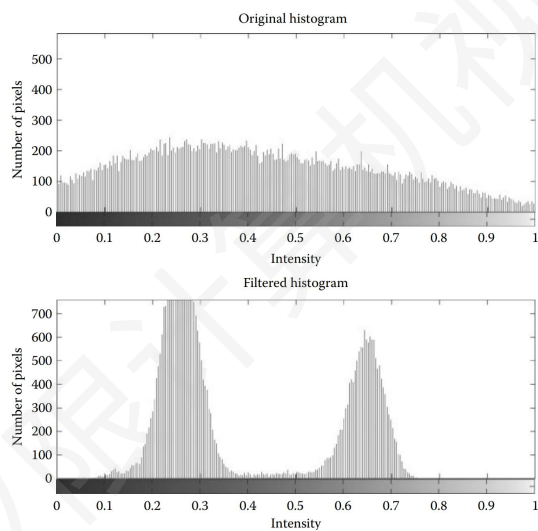


$\text{blur}_x(I)$

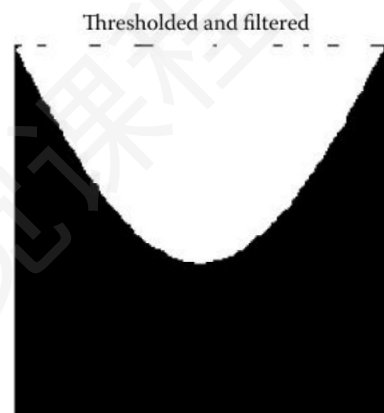


$\text{blur}_y(\text{blur}_x(I))$

规格严格 功夫到家



Filtered figure



Thresholded and filtered

规格严格 功夫到家

双边滤波

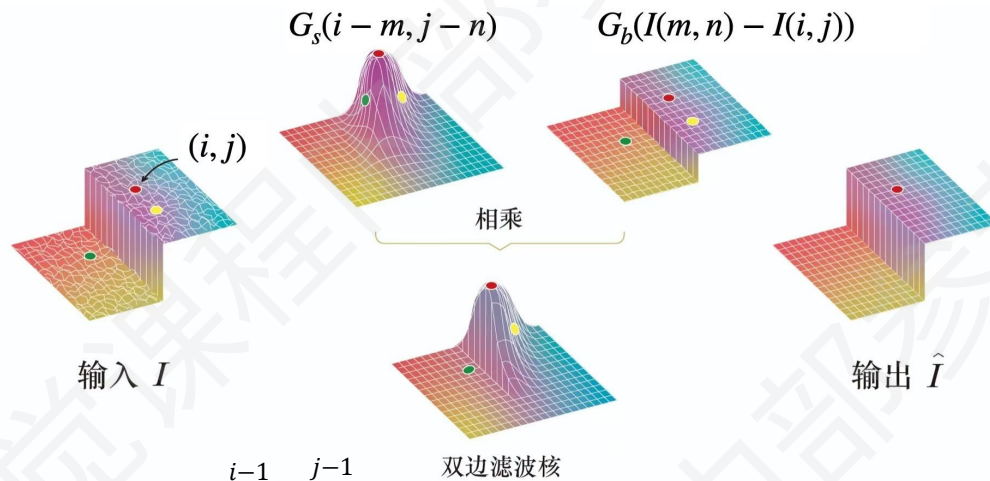
規格嚴格 功夫到家

Bilateral filtering (双边滤波)

- What if we were to combine the idea of a weighted filter kernel with a better version of outlier rejection?
- What if instead of rejecting a fixed percentage α , we simply reject (in a soft way) pixels whose values differ too much from the central pixel value?
- In the bilateral filter, the output pixel value depends on a weighted combination of neighboring pixel values!

規格嚴格 功夫到家

Bilateral filtering (双边滤波)



$$\hat{I}(i, j) = \frac{1}{W_{sb}} \sum_{m=i-k}^{i-1} \sum_{n=j-k}^{j-1} I(m, n) G_s(i-m, j-n) G_b(I(m, n) - I(i, j))$$

规格严格 功夫到家

Bilateral filtering

- Domain Kernel

$$d(i, j, k, l) = \exp \left(-\frac{(i-k)^2 + (j-l)^2}{2\sigma_d^2} \right)$$
- Range Kernel

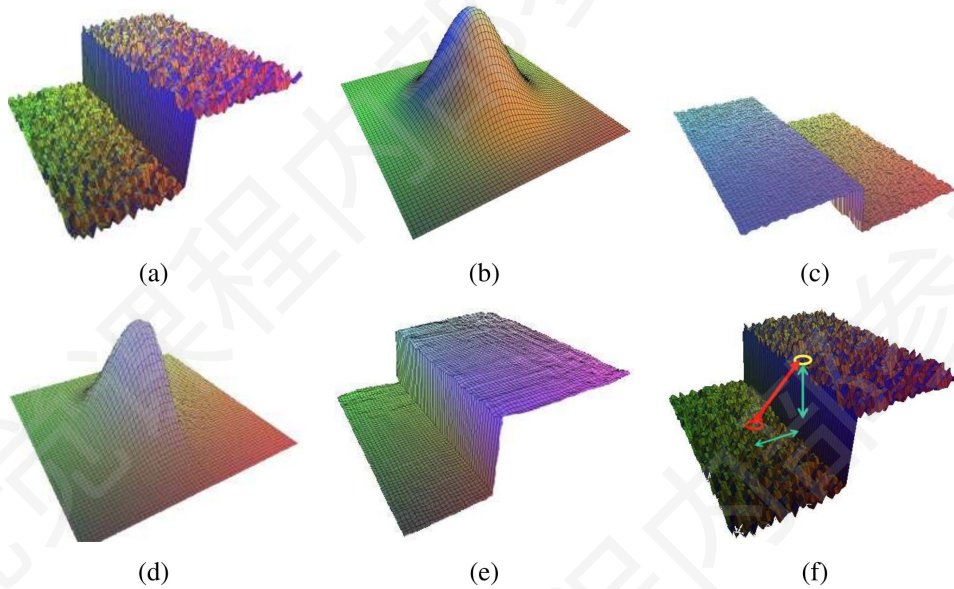
$$r(i, j, k, l) = \exp \left(-\frac{\|\mathbf{f}(i, j) - \mathbf{f}(k, l)\|^2}{2\sigma_r^2} \right)$$
- Product of Domain Kernel and Range Kernel

$$w(i, j, k, l) = \exp \left(-\frac{(i-k)^2 + (j-l)^2}{2\sigma_d^2} - \frac{\|\mathbf{f}(i, j) - \mathbf{f}(k, l)\|^2}{2\sigma_r^2} \right)$$
- The output pixel value

$$\mathbf{g}(i, j) = \frac{\sum_{k,l} \mathbf{f}(k, l) w(i, j, k, l)}{\sum_{k,l} w(i, j, k, l)}$$

规格严格 功夫到家

Bilateral filtering

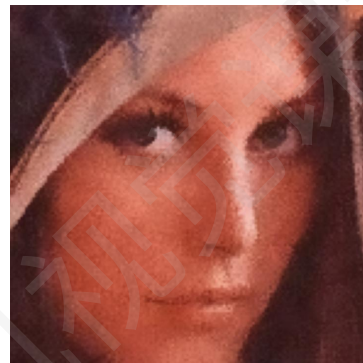
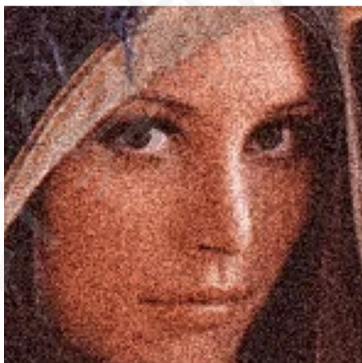


規格嚴格 功夫到家

Bilateral filtering (双边滤波)

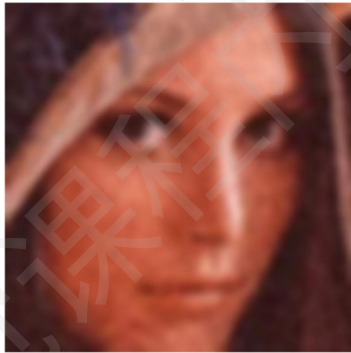
$$G_s(m, n) = \frac{1}{2\pi\sigma_s^2} e^{-\frac{m^2 + n^2}{2\sigma_s^2}}$$

$$G_b(d) = \frac{1}{2\pi\sigma_b^2} e^{-\frac{d^2}{2\sigma_b^2}}$$

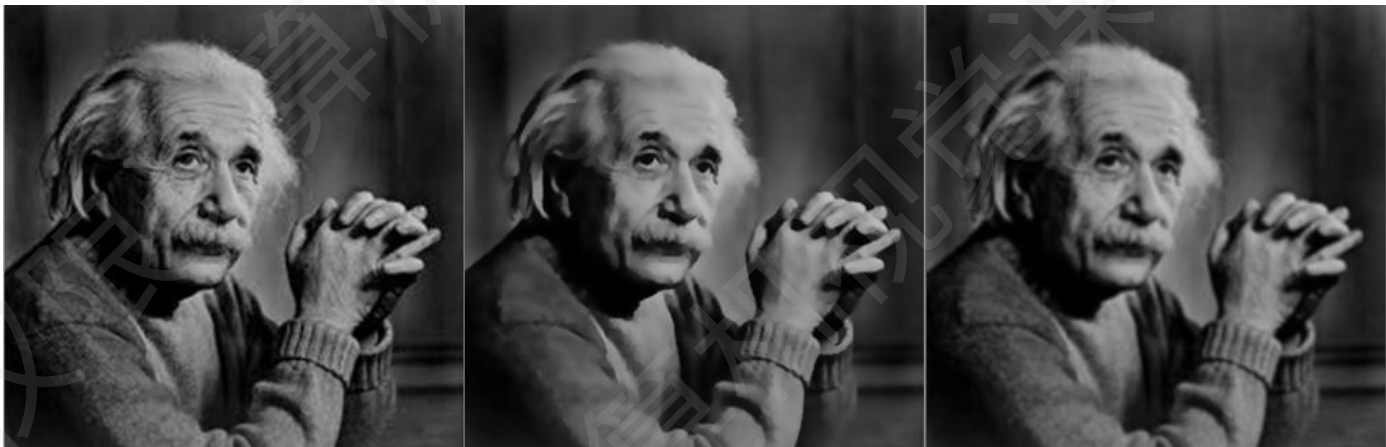
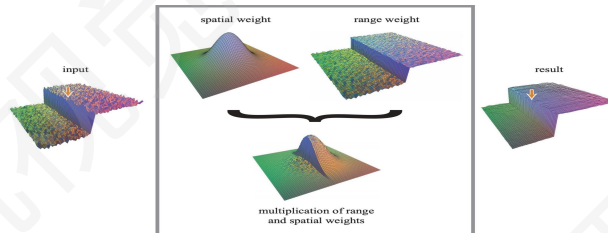


$$\sigma_s = 10, \sigma_b = 110$$

規格嚴格 功夫到家



規格嚴格 功夫到家

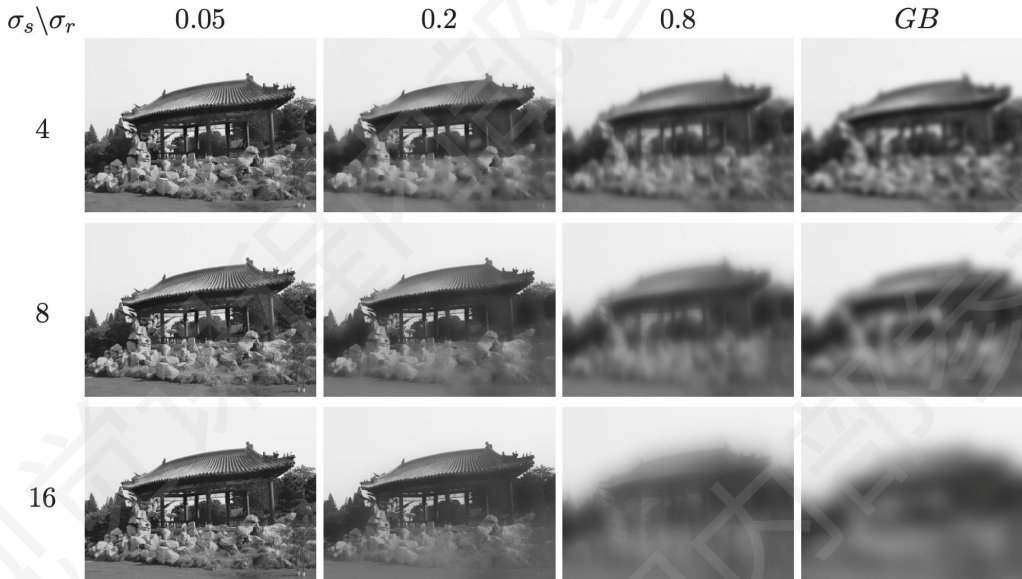


規格嚴格 功夫到家



哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY

Bilateral filter



[Paris et al. "A Gentle Introduction to Bilateral Filtering and its Applications", 2008]

規格嚴格 功夫到家



哈爾濱工業大學
HARBIN INSTITUTE OF TECHNOLOGY

Denoising with Gaussian



Noisy
image



Gaussian blur = 3



Gaussian blur = 5

Source: [Paris et al. 2008]

規格嚴格 功夫到家



哈尔滨工业大学
HARBIN INSTITUTE OF TECHNOLOGY

Denoising with Bilateral Filter



Original



Noisy
image



Bilateral
filtering

Source: [Paris et al, 2008]
规格严格 功夫到家

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哈尔滨工业大学
HARBIN INSTITUTE OF TECHNOLOGY

Iteratively applying the filter



1 iteration



2 iteration



4 iteration

Source: [Paris et al, 2008]
规格严格 功夫到家

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Iteratively applying the filter



“Cartoonifying” a
video

[Winnermoller et al., 2006]

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Any Improvements?



Guidance I



Filter Input p

規格嚴格 功夫到家

- The guided image is used to compute the locally adapted inter-pixel weights

$$\mathbf{g}(i, j) = \sum_{k, l} w(\mathbf{h}; i, j, k, l) \mathbf{f}(k, l)$$

- Modeling the relationship between the guide and input images using a local affine transformation

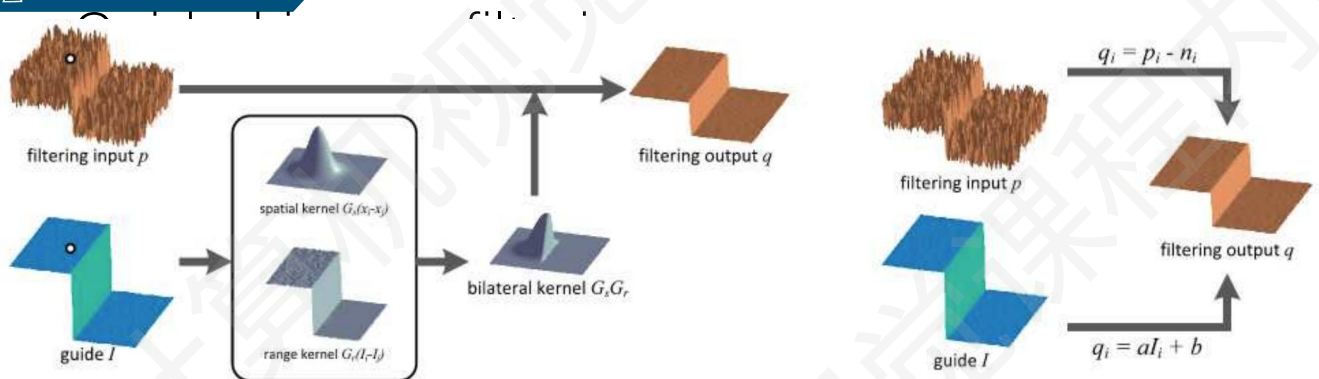
$$\mathbf{g}(i, j) = \mathbf{A}_{k, l} \mathbf{h}(i, j) + \mathbf{b}_{k, l}$$

- Ridge regression

- The estimates for $\mathbf{A}_{k, l}$ and $\mathbf{b}_{k, l}$ are obtained from a regularized least squares fit over a square neigh

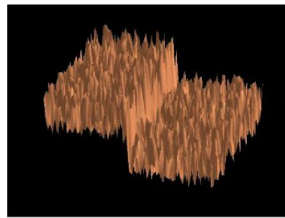
$$\sum_{(i, j) \in \mathcal{N}_{k, l}} \|\mathbf{A}_{k, l} \mathbf{h}(i, j) + \mathbf{b}_{k, l} - \mathbf{f}(i, j)\|^2 + \lambda \|\mathbf{A}\|^2$$

規格嚴格 功夫到家

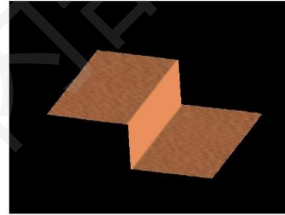
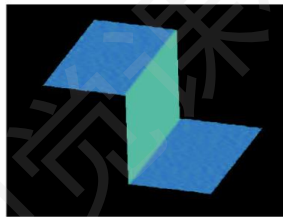


- Unlike joint bilateral filtering, shown on the left, GIF computes a per pixel weight mask from the guide image (shown as I in the figure), the guided image filter models the output value (shown as q_i in the figure) as a local affine transformation of the guide pixels.

規格嚴格 功夫到家

Input p

$$q_i = p_i - n_i$$

 n_i - noise / texture
Output q Guide I

$$\nabla q_i = a \nabla I_i$$

$$q_i = a I_i + b$$

Bilateral/joint bilateral filter
does not have this linear model

For each local window

$$p_i \approx a I_i + b$$

$$\min_{a,b} \sum_i (a I_i + b - p_i)^2 + \epsilon a^2$$

Linear regression

 $O(N^2) \rightarrow O(N)$

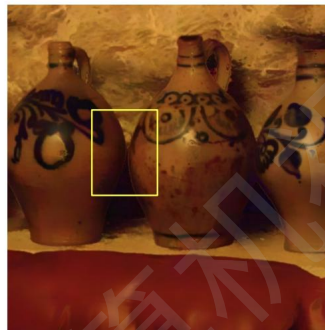
$$a = \frac{\text{cov}(I, p)}{\text{var}(I) + \epsilon}$$

$$b = \bar{p} - a \bar{I}$$

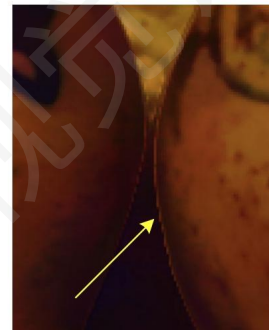
规格严格 功夫到家

Guidance I 

Guided Filter

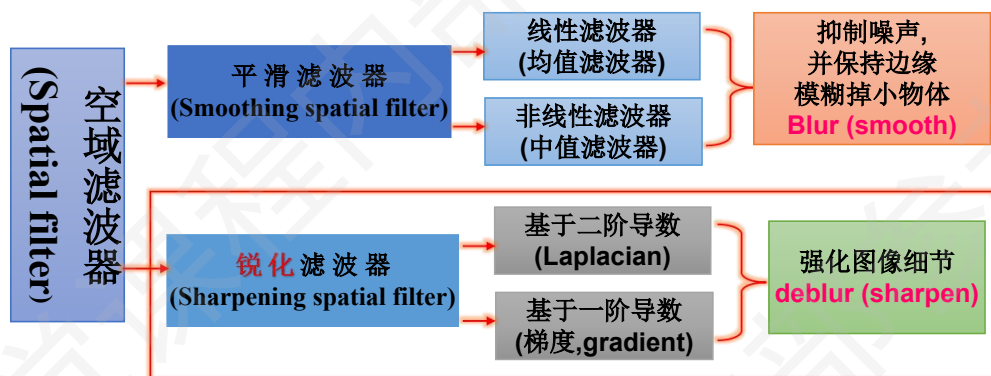
Filter Input p 

Joint Bilateral Filter



规格严格 功夫到家

空间滤波器



规格严格 功夫到家

锐化

- 边缘和微分滤波器

规格严格 功夫到家

锐化滤波器



规格严格 功夫到家

锐化滤波器

➤ 锐化滤波器的主要应用：

- 突出图像中的**细节**并增强模糊的细节。
- **强调**印刷中的**细微层次**并补偿扫描过程中图像的模糊。
- 通过锐化**改善**超声成像中的**低分辨率和边缘模糊**。
- 在图像识别中，在**分割之前增强边缘**，以便于后续提取。
- 通过锐化处理恢复过度模糊或曝光不足的图像。
- 先进武器系统中的**目标识别与定位**。

规格严格 功夫到家

如何锐化图像？

規格嚴格 功夫到家

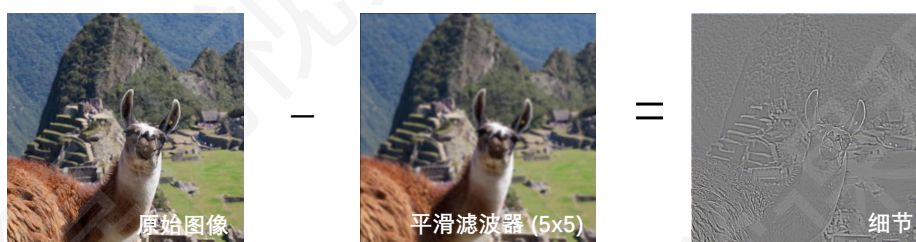
3.1 滤波

3.2 卷积

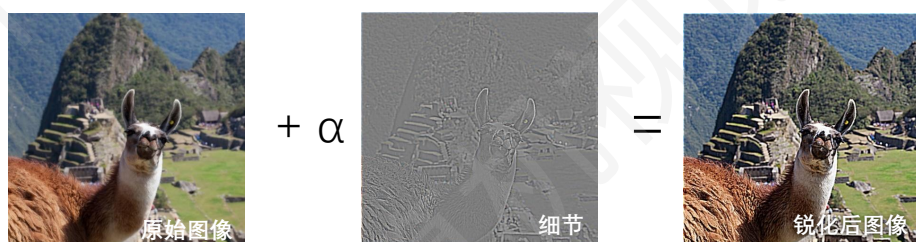
3.3 边缘检测

锐化滤波器

➤ 可以通过平滑滤波器实现么？



让我们把细节加回来：



規格嚴格 功夫到家

锐化滤波器

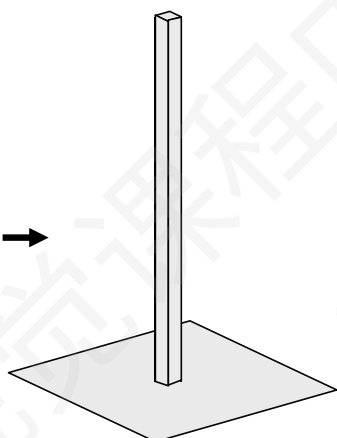
$$F + \alpha (F - F * H) =$$

$\uparrow$ $\underbrace{\hspace{2cm}}$
 原始图像 模糊图像

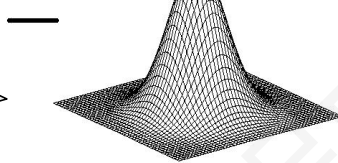
图像卷积运算律:

线性性: $I * (aK_1 + bK_2) = a(I * K_1) + b(I * K_2)$

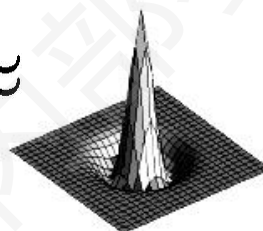
单位脉冲
(中心为
1, 其余
为0的
恒等核)



缩放脉冲



高斯滤波器



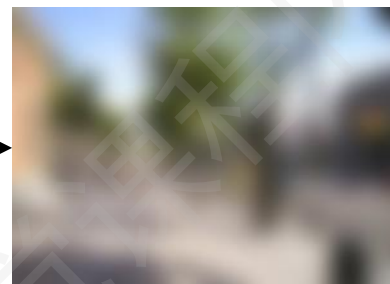
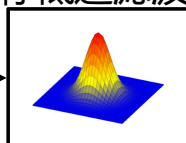
锐化滤波器

格 功夫到家

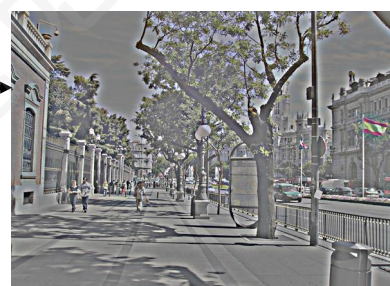
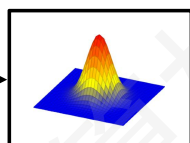
图像分解



高斯滤波器
(又称低通滤波器)

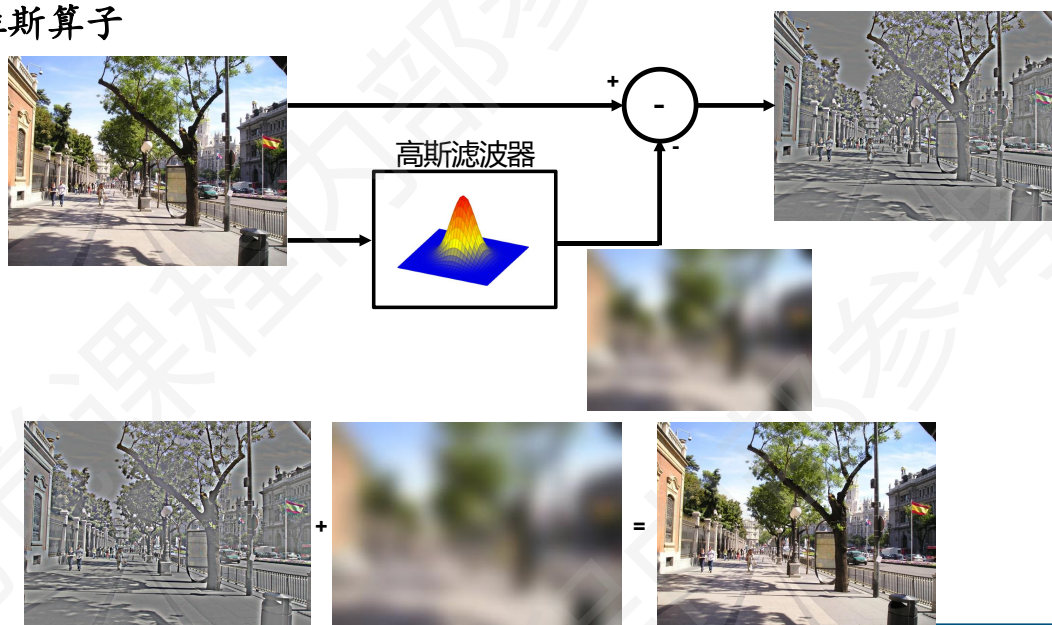


近似拉普拉斯算子
(又称高通滤波器)



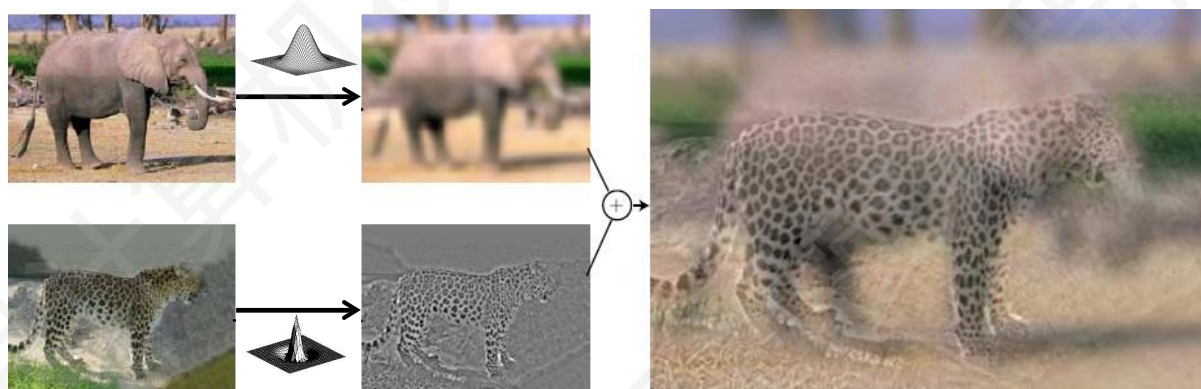
规格严格 功夫到家

近似拉普拉斯算子



规格严格 功夫到家

Application: Hybrid Images



[A. Oliva, A. Torralba, P.G. Schyns, [Hybrid Images](#), SIGGRAPH 2006]

规格严格 功夫到家

微分濾波器

規格嚴格 功夫到家

微分濾波器

➤ 原理

$$\left. \frac{\partial f(x)}{\partial x} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

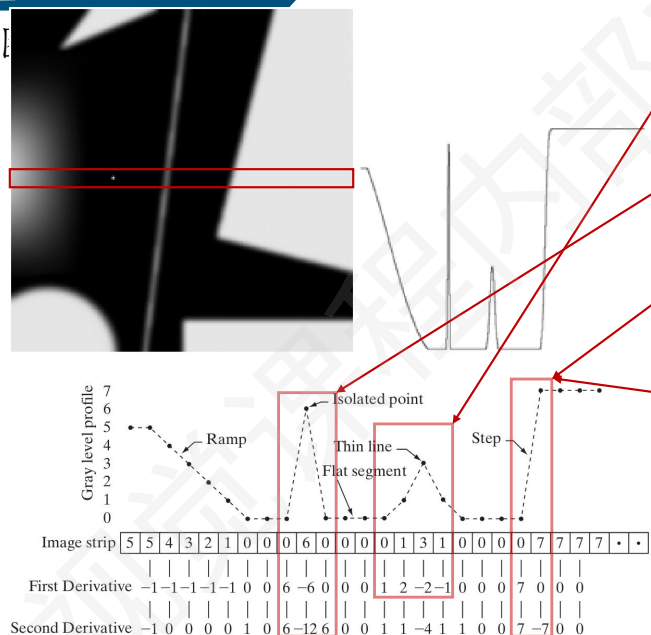
- 圖像處理中應用微分最常見的方法是計算梯度。
- 函數 $f(x, y)$ 在點 (x, y) 處的一階和二階微分定義如下：

$\Delta x = 1$

一階導數近似：
$$\frac{\partial f(x)}{\partial x} = f(x+1) - f(x)$$

二階導數近似：
$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} &= f'(x) - f'(x-1) \\ &= [f(x+1) - f(x)] - [f(x) - f(x-1)] \\ &= f(x+1) + f(x-1) - 2f(x) \end{aligned}$$

規格嚴格 功夫到家



- 一阶微分通常会导致边缘（非跳变）变粗。
- 二阶微分对细节具有更强的响应。
- 一阶微分对灰度阶跃具有较强的响应。
- 二阶微分对灰度阶跃变化产生双线响应。
- 当灰度变化相似时，二阶微分响应的幅值大小关系为：点 > 线 > 阶跃。

規格嚴格 功夫到家



数学基础

图像在 (i,j) 处的一阶微分 (又称**梯度**) :

$$\nabla I(i, j) = [I_x(i, j), I_y(i, j)]$$

图像的一阶微分是通过一阶差分实现的。

$$I_x(i, j) = \frac{1}{2} \sum_{m=-1}^1 \sum_{n=-1}^1 I(i-m, j-n) K_x(m, n)$$

等价于卷积形式:

$$\mathbf{I}_x = \mathbf{I} * \mathbf{K}_x$$

$$I_y(i, j) = \frac{1}{2} \sum_{m=-1}^1 \sum_{n=-1}^1 I(i-m, j-n) K_y(m, n)$$

$$\mathbf{I}_v = \mathbf{I} * \mathbf{K}_v$$

$$K_x = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad K_y = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

規格嚴格 功夫到家²

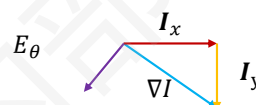
数学基础

➤ 图像在 (i,j) 处的边缘强度

$$E_s(i, j | I) = \|\nabla I(i, j)\| = \sqrt{I_x^2(i, j) + I_y^2(i, j)}$$

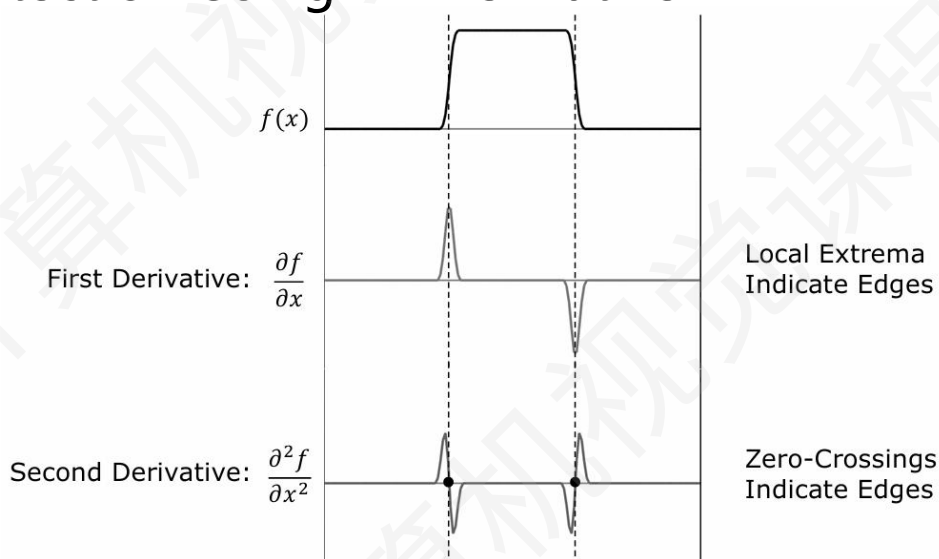
➤ 图像在 (i,j) 处的边缘方向（与梯度方向垂直）

$$E_\theta(i, j | I) = \tan^{-1} \frac{I_y(i, j)}{I_x(i, j)} + \frac{\pi}{2}$$



规格严格 功夫到家

Edge Detection Using 2nd Derivative



规格严格 功夫到家

锐化滤波器

- 一阶微分滤波器 – 梯度算子
- 二阶微分滤波器 – 拉普拉斯算子

規格嚴格 功夫到家

一阶微分

規格嚴格 功夫到家

一阶微分

输入图像

$$f(x, y)$$

与梯度算子卷积
(Sobel算子等)

增强图像

$$g(x, y) = \nabla f(x, y)$$

- 利用梯度增强图像边缘（用于后续边缘检测）

非旋转不变的

梯度是一个矢量

$$\nabla \mathbf{f} = \left[\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \right]^T$$

旋转不变的

非线性的

通常使用梯度的大小来代替梯度本身

$$\nabla f = \text{mag}(\nabla \mathbf{f}) = \left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]^{1/2}$$

通常使用绝对值而不是平方值

$$\nabla f = \text{mag}(\nabla \mathbf{f}) \approx \left| \frac{\partial f}{\partial x} \right| + \left| \frac{\partial f}{\partial y} \right|$$

近似的

规格严格 功夫到家

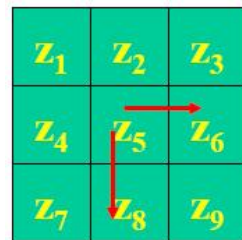
一阶微分算子

- 考虑一个 3x3 的图像区域，其中 z 表示灰度级。点 z_5 处的 ∇f 值可以通过对 $z_6 - z_5$ 和 $z_8 - z_5$ 进行数值近似，它们可以组合为：

$$\nabla f \approx \left[(z_6 - z_5)^2 + (z_8 - z_5)^2 \right]^{1/2} \quad G_x = \frac{\partial f}{\partial x} \quad G_y = \frac{\partial f}{\partial y}$$

简化为：

$$\nabla f \approx |z_6 - z_5| + |z_8 - z_5|$$



规格严格 功夫到家

1st order differentiation operator

➤ Differential filter template coefficient design

➤ Roberts cross gradient operator

$$\nabla f \approx |z_9 - z_5| + |z_8 - z_6|$$

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

- ✓ The gradient calculation consists of **two templates**, the first one obtains the first term of the gradient, the second one obtains the second term of the gradient, and then they are **summed to obtain the gradient**.
- ✓ The two templates are called Roberts cross gradient operator.

-1	0
0	1

0	-1
1	0

规格严格 功夫到家

1st order differentiation operator

➤ Differential filter template coefficient design

➤ Prewitt gradient operator

➤ A 3x3 template

$$\nabla f \approx |(z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)| + |(z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)|$$

-1	-1	-1
0	0	0
1	1	1

-1	0	1
-1	0	1
-1	0	1

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

规格严格 功夫到家

1st order differentiation operator

➤ Differential filter template coefficient design

➤ Sobel 索贝尔 gradient operator

➤ A 3x3 template $\nabla f \approx \left| (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3) \right| + \left| (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7) \right|$

➤ 2 is the weight that enhances the importance of the center pixel.

-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

规格严格 功夫到家

一阶微分算子

➤ 差分滤波器模板系数设计

➤ Sobel 索贝尔 梯度算子

➤ 3x3 模板 $\nabla f \approx \left| (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3) \right| + \left| (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7) \right|$

➤ 2 是用于增强中心像素重要性的权重。

-1	-2	-1
0	0	0
1	2	1

垂直索贝尔滤波器

-1	0	1
-2	0	2
-1	0	1

水平索贝尔滤波器

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

规格严格 功夫到家

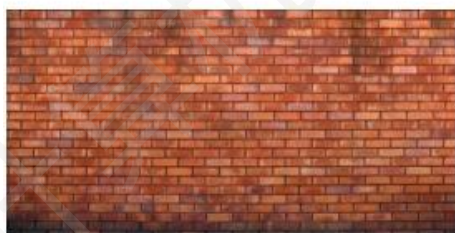
Sobel 是最受青睐的梯度算子，几乎是传统图像梯度计算的“标配”默认选项。

索贝尔(Sobel)算子：示例

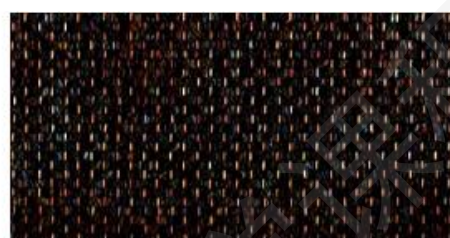


規格嚴格 功夫到家

索贝尔(Sobel)算子：示例



original



horizontal Sobel filter

$$\frac{1}{8} \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

s_x



vertical Sobel filter

$$\frac{1}{8} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

s_y

規格嚴格 功夫到家

索贝尔 (Sobel) 滤波器

$$K_x = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} \quad K_y = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

$$K_x = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} +1 & 0 & -1 \end{bmatrix}$$

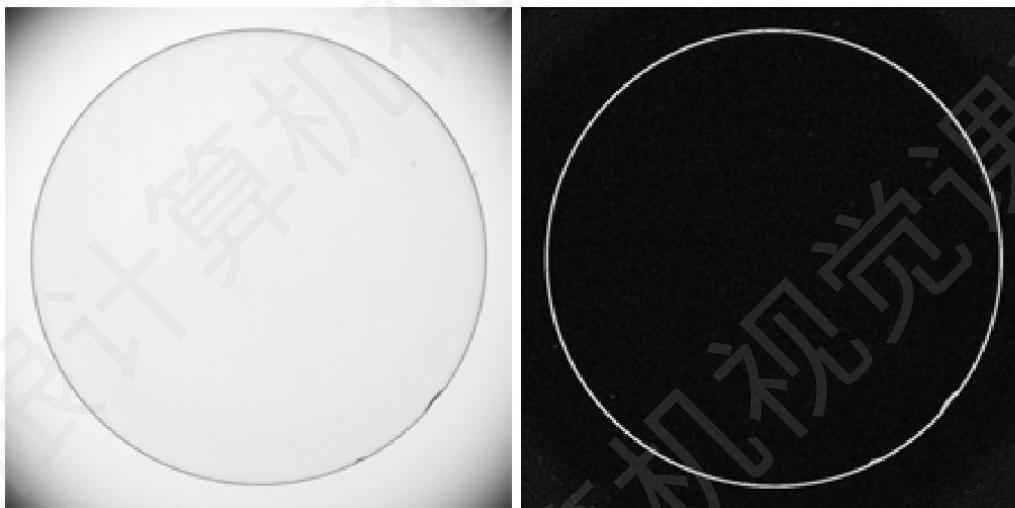
Similar to the effect of Gaussian smoothing

After obtaining the gradients along the X-axis and Y-axis using the Sobel operator, the image gradient magnitude is calculated using an approximation without square root.

$$E_s(i, j | I) = \|\nabla I(i, j)\| \approx \|I_x(i, j)\| + \|I_y(i, j)\|$$

規格嚴格 功夫到家

1st order differentiation operator



a b

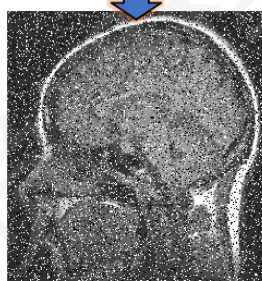
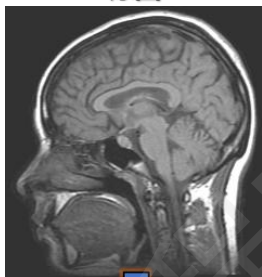
FIGURE 3.45

Optical image of contact lens (note defects on the boundary at 4 and 5 o'clock).
 (b) Sobel gradient.
 (Original image courtesy of Mr. Pete Sites, Perceptics Corporation.)

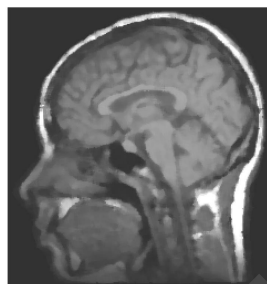
規格嚴格 功夫到家

部分锐化方法效果的比较

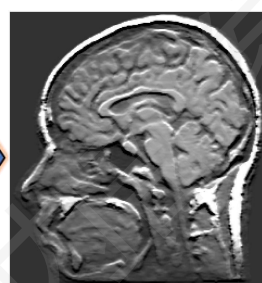
原图



概率为0.2的椒盐
噪声污染



经5×5的模板
中值滤波



3×3 sobel 滤波

规格严格 功夫到家

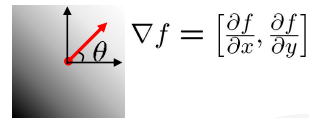
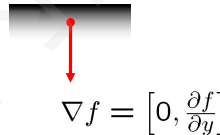
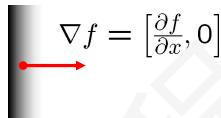
图像的梯度

规格严格 功夫到家

Image gradient

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$

The gradient points in the direction of most rapid increase in intensity



The *edge strength* is given by the gradient **magnitude**:

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

The gradient **direction** is given by:

$$\theta = \tan^{-1} \left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x} \right)$$

- How does this relate to the direction of the edge?

規格嚴格 功夫到家

Gradient	Roberts	Prewitt	Sobel (3x3)	Sobel (5x5)
$\frac{\partial I}{\partial x}$	$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & -2 & 0 & 2 & 1 \\ -2 & -3 & 0 & 3 & 2 \\ -3 & -5 & 0 & 5 & 3 \\ -2 & -3 & 0 & 3 & 2 \\ -1 & -2 & 0 & 2 & 1 \end{bmatrix}$
$\frac{\partial I}{\partial y}$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ -1 & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 2 & 3 & 5 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 2 & 1 \end{bmatrix}$

Good Localization
Noise Sensitive
Poor Detection

Poor Localization
Less Noise Sensitive
Good Detection

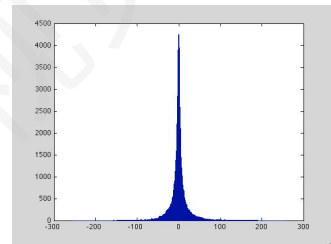
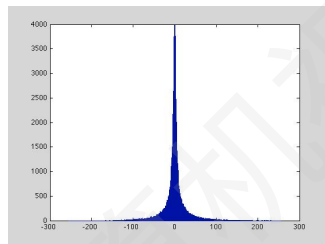
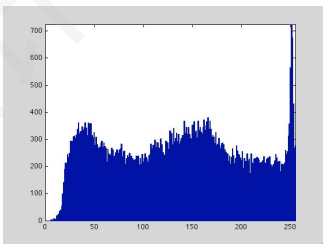
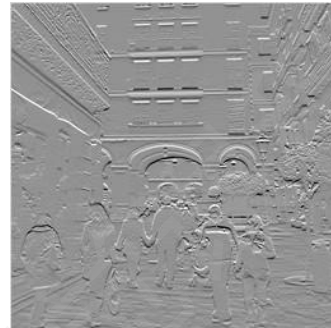
規格嚴格 功夫到家

Other Uses of Image Gradient

- Image statistics
- Gradient histogram

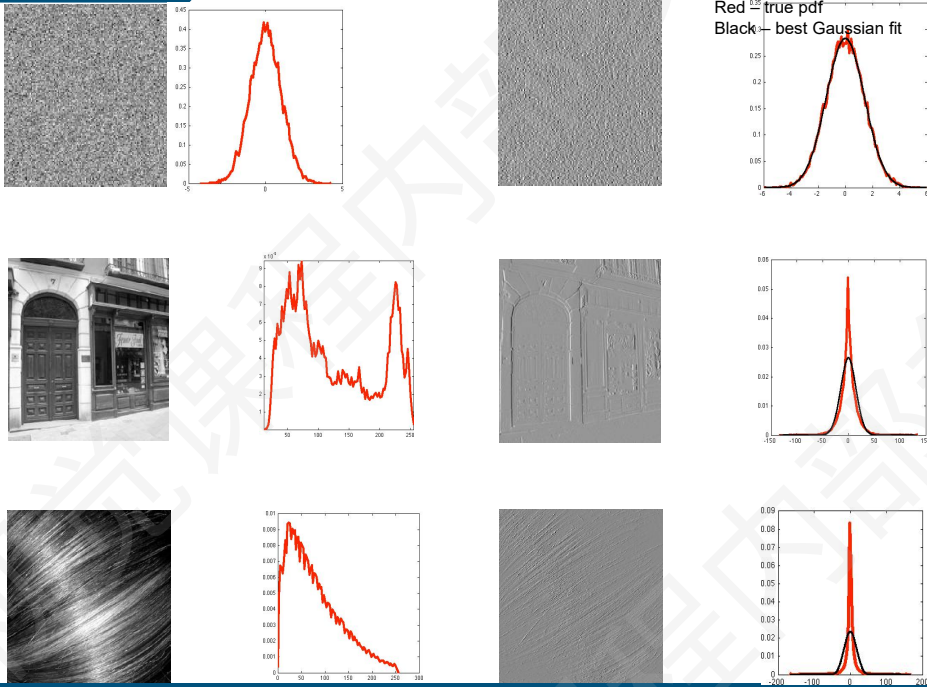
規格嚴格 功夫到家

Other response distribution is pretty consistent!



規格嚴格 功夫到家

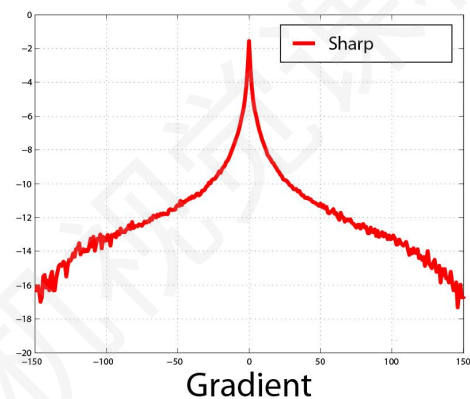
Intensity histogram [1 -1] filter output [1 -1] output histogram



Natural image statistics

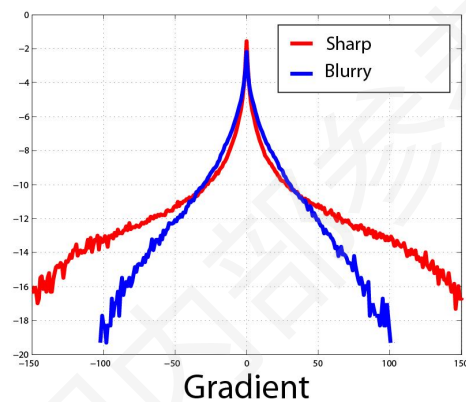
Characteristic distribution with heavy tails

Histogram of image gradients



Blurry images have different statistics

Histogram of image gradients



規格嚴格 功夫到家
Source: R. Fergus

二阶微分

規格嚴格 功夫到家



Recap: 微分濾波器

► 原理

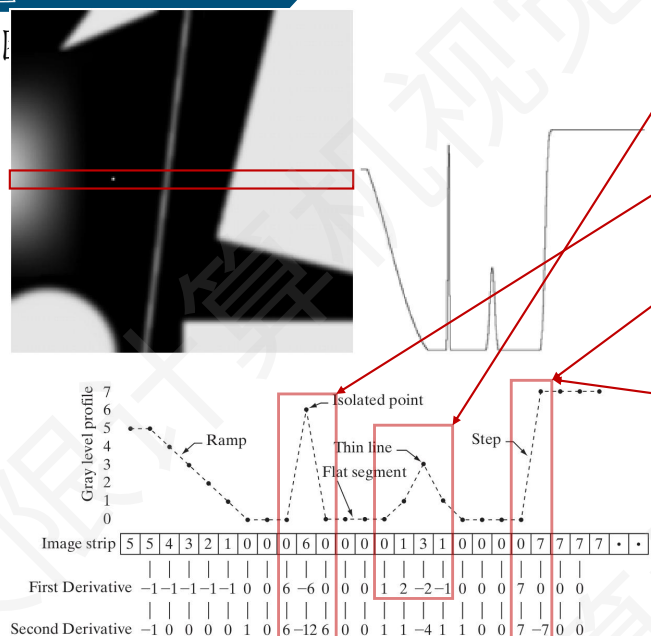
$$\left. \frac{\partial f(x)}{\partial x} \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

- 图像处理中应用微分最常见的方法是计算梯度。
- 函数 $f(x, y)$ 在点 (x, y) 处的一阶和二阶微分定义如下:

一阶导数近似: $\frac{\partial f(x)}{\partial x} = f(x+1) - f(x)$

二阶导数近似: $\frac{\partial^2 f}{\partial x^2} = f'(x) - f'(x-1)$
 $= [f(x+1) - f(x)] - [f(x) - f(x-1)]$
 $= f(x+1) + f(x-1) - 2f(x)$

規格嚴格 功夫到家



- 一阶微分通常会导致边缘（非跳变）变粗。
- 二阶微分对细节具有更强的响应。
- 一阶微分对灰度阶跃具有较强的响应。
- 二阶微分对灰度阶跃变化产生双线响应。
- 当灰度变化相似时，二阶微分响应的幅值大小关系为：点 > 线 > 阶跃。

規格嚴格 功夫到家

二阶微分滤波器 – 拉普拉斯算子

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y)$$

			$f(x-1, y+1)$		$f(x+1, y+1)$
0	$f(x, y+1)$	0	1	1	1
$f(x-1, y)$	$f(x, y)$	$f(x+1, y)$	1	-8	1
0	$f(x, y-1)$	0	1	1	1
			$f(x-1, y-1)$		$f(x+1, y-1)$
0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

規格嚴格 功夫到家

二阶微分滤波器 – 拉普拉斯算子

0	1	0
1	-4	1
0	1	0

对角拓展

1	1	1
1	-8	1
1	1	1

$$\nabla^2 f = \begin{matrix} \text{水平方向} \\ \text{二阶偏导} \end{matrix} + \begin{matrix} \text{垂直方向} \\ \text{二阶偏导} \end{matrix} \quad \nabla^2 f = \begin{matrix} \text{水平方向} \\ \text{二阶偏导} \end{matrix} + \begin{matrix} \text{垂直方向} \\ \text{二阶偏导} \end{matrix} + \begin{matrix} 45^\circ \text{方向} \\ \text{二阶偏导} \end{matrix} + \begin{matrix} 135^\circ \text{方向} \\ \text{二阶偏导} \end{matrix}$$

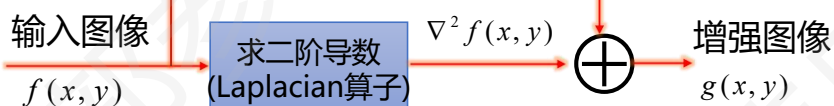
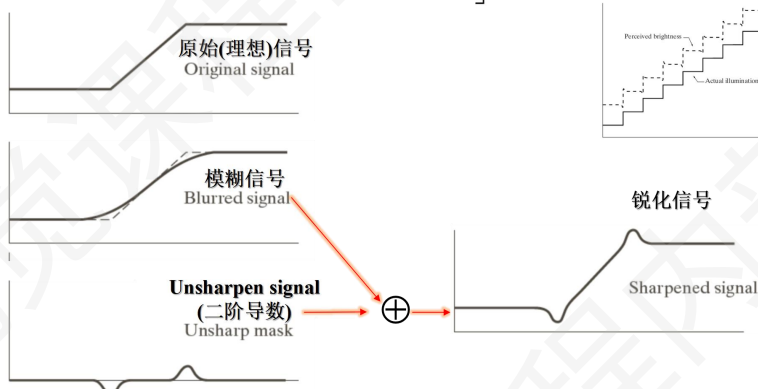
◆ **滤波各向同性**：各向同性滤波器具有**旋转不变性**，即滤波响应基本不受图像灰度变化方向的影响。图像在旋转前后经滤波所得结果相同。

◆ 为考虑 45°各向同性，可在 90°基准之上叠加对角滤波，生成右侧模板。

規格嚴格 功夫到家

基于二阶微分的增强

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) & (1) \\ f(x, y) + \nabla^2 f(x, y) & (2) \end{cases}$$



规格严格 功夫到家

增强拉普拉斯滤波器的结果



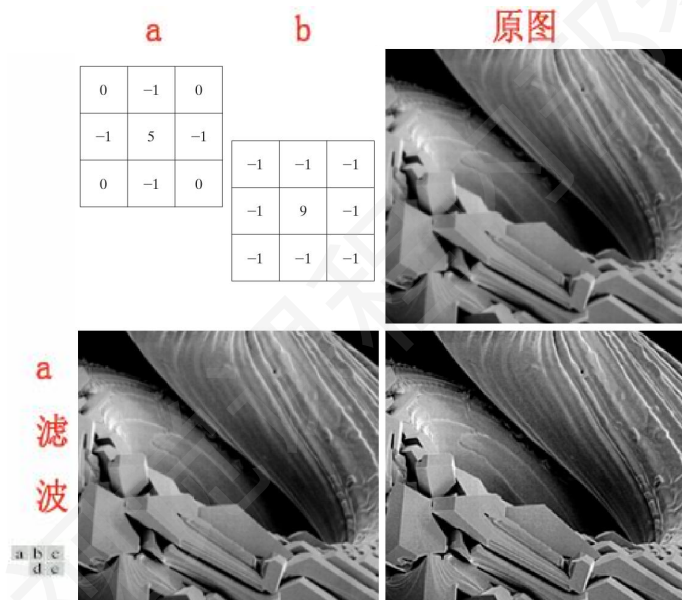
原图

中心值为-4的拉普拉斯滤波

中心值为-8的拉普拉斯滤波

规格严格 功夫到家

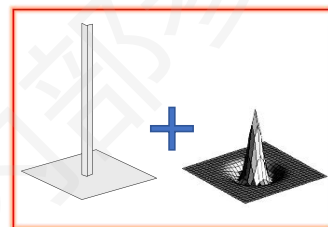
增强拉普拉斯滤波器的结果



$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) & (1) \\ f(x, y) + \nabla^2 f(x, y) & (2) \end{cases}$$

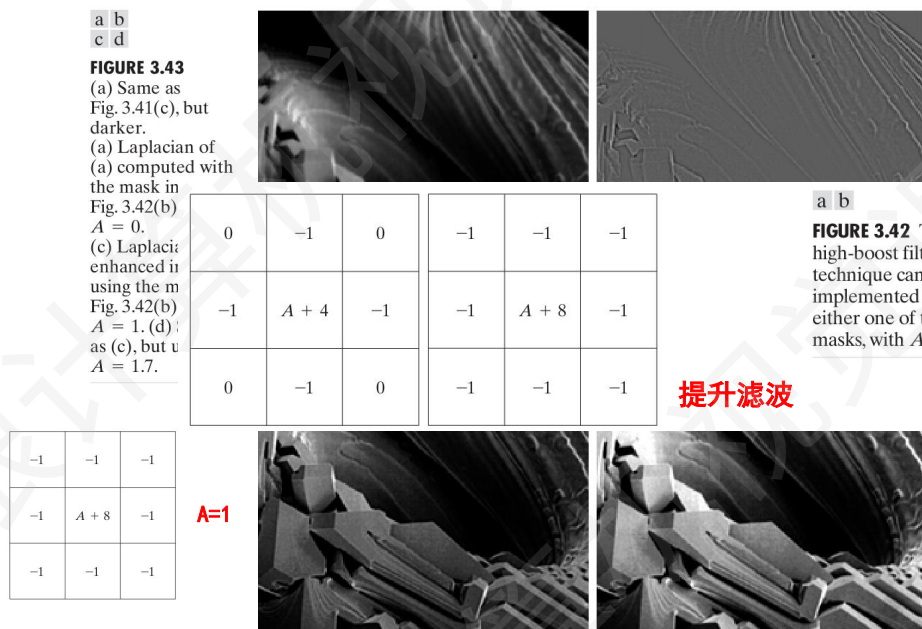
通常直接把前面的两个式子合到一起，从而产生一个单一的掩模，进而只需要用掩模对图像进行一次扫描即可实现变换

比
a 更
b 锐
化



规格严格 功夫到家

增强拉普拉斯滤波器的结果

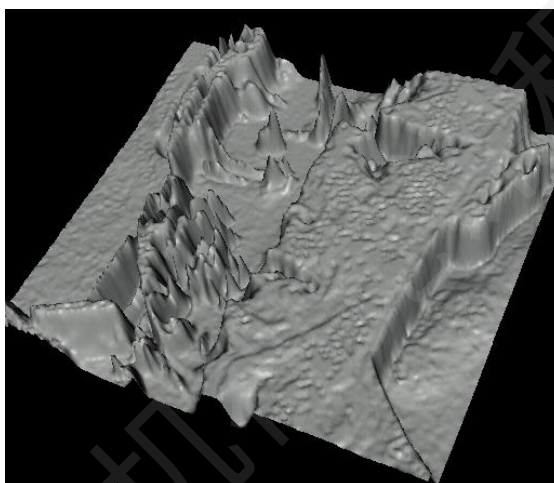


规格严格 功夫到家

边缘 Edges

規格嚴格 功夫到家

Recap: Images as functions...

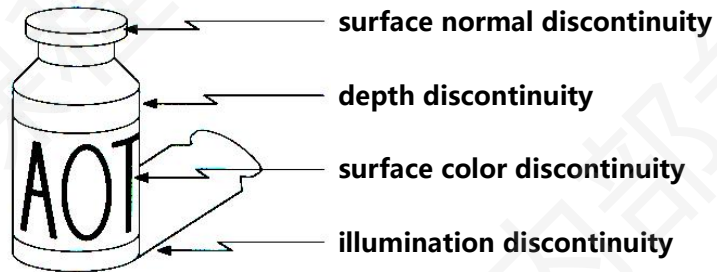


- Edges look like steep cliffs

規格嚴格 功夫到家

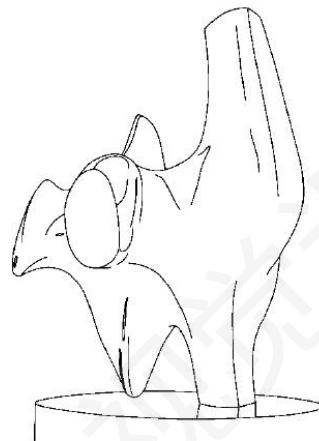
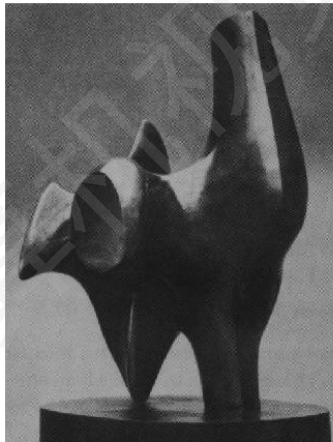
Origin of edges

- Edges are caused by a variety of factors



規格嚴格 功夫到家

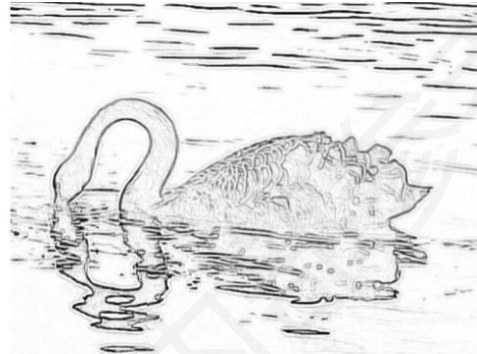
Edge c



- Convert a 2D image into a set of curves
 - Extracts **salient features of the scene**
 - **More compact than pixels**

規格嚴格 功夫到家

Edges are locations in an image where brightness, texture, or color change sharply, containing information about object shapes and scene structures.



Edge detection is the foundation of visual information processing

規格嚴格 功夫到家

Recap: edges are more compact than pixels



- Does the human eye need all pixels within a frame ($m \times n$ pixels) to perceive the content of an image?
- Is the human eye's perception of image content affected by pixel density?

哈哈

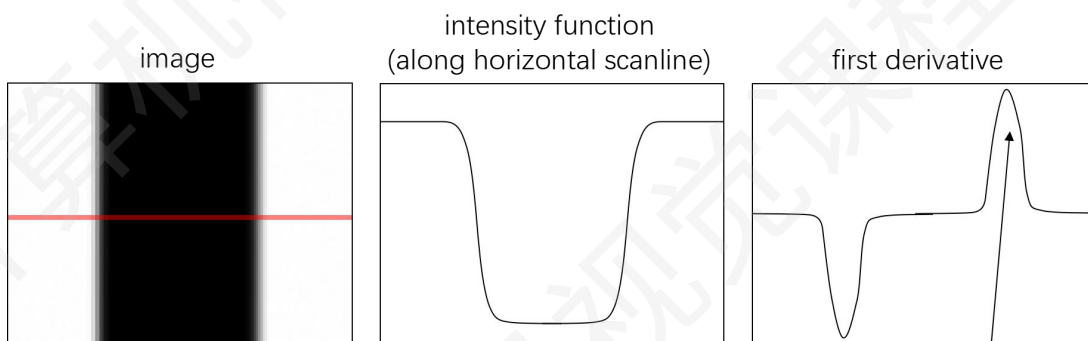
規格嚴格 功夫到家

如何检测边缘？

規格嚴格 功夫到家

如何检测边缘？

- 如何检测图像中的边缘（例如函数中的不连续点）？
- 求导：导数在不连续点处很大。

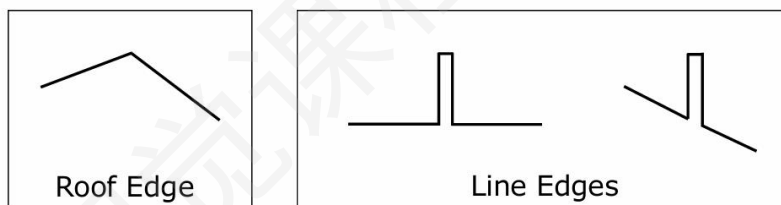
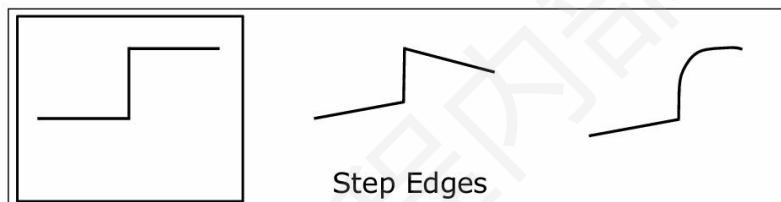


- 边缘是图像强度函数快速变化的区域。

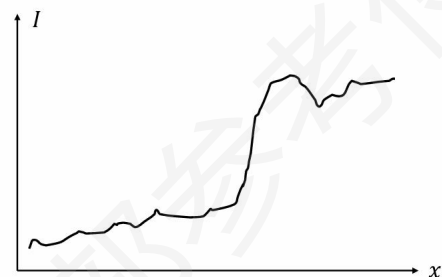
edges correspond to
extrema of derivative

規格嚴格 功夫到家

边缘类型



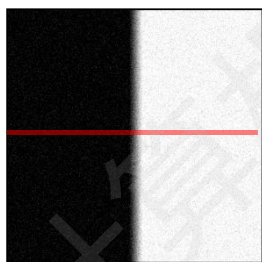
真实边缘



问题：噪声图像和离散图像

规格严格 功夫到家

噪声的影响



带噪输入图像

first-order
finite difference $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+0.5h) - f(x-0.5h)}{h}$

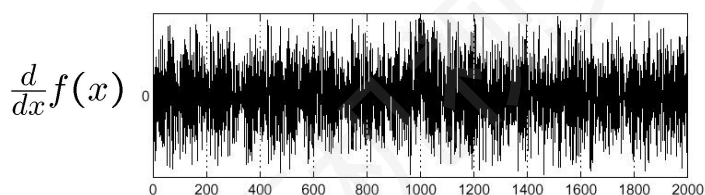
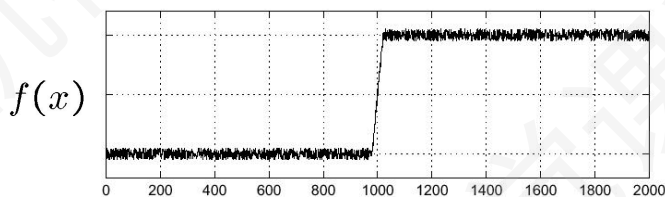
1D derivative filter

$$\begin{bmatrix} 1 & 0 & -1 \end{bmatrix}$$

second-order
finite difference $f''(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$

Laplace filter

$$\begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$$

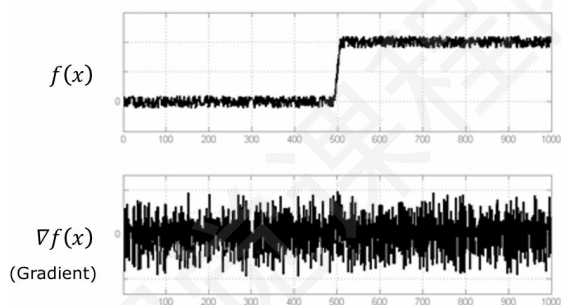


Source: S. Seitz

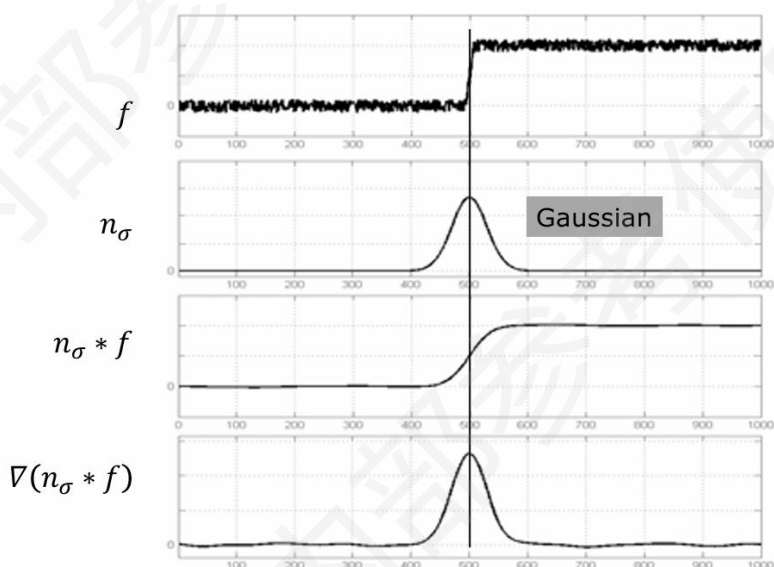
边缘在哪里？

如何解决？

规格严格 功夫到家



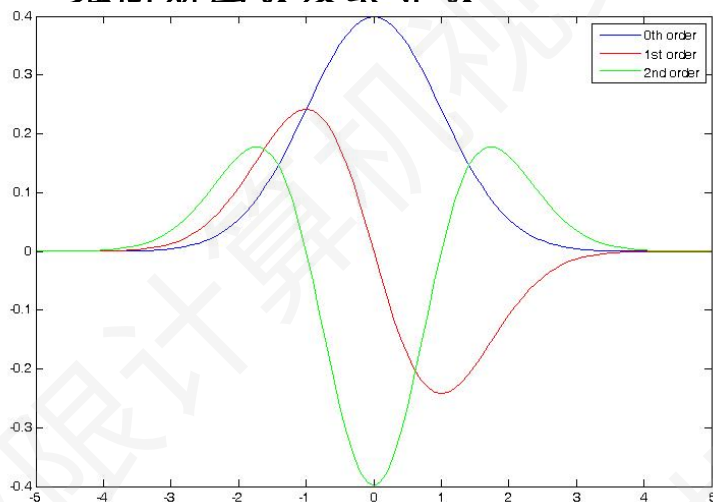
噪声的影响



解决方案：先进行高斯平滑

规格严格 功夫到家

一维高斯函数及其导数



$$G_\sigma(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$$

$$G'_\sigma(x) = -\frac{x}{\sigma^2} G_\sigma(x)$$

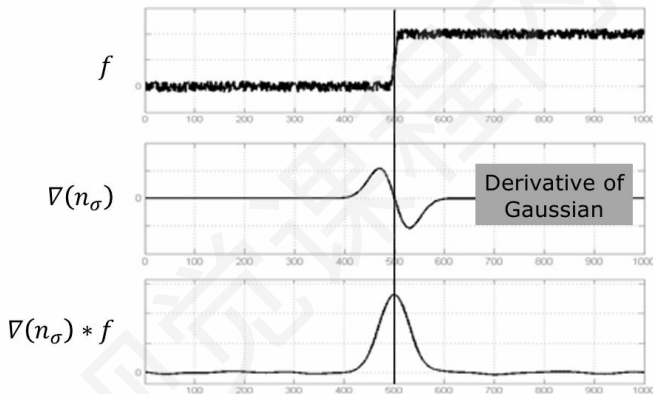
$$G''_\sigma(x) = \left(\frac{x^2}{\sigma^4} - \frac{1}{\sigma^2} \right) G_\sigma(x)$$

- 可以通过先用高斯滤波器平滑图像，然后取一阶或二阶导数来创建更复杂的核

规格严格 功夫到家

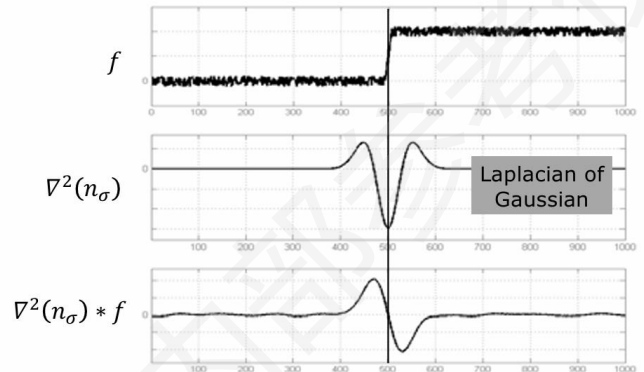
Derivative of Gaussian ($\nabla(n_\sigma)$)

$\nabla(n_\sigma * f) = \nabla(n_\sigma) * f$...saves us one operation.



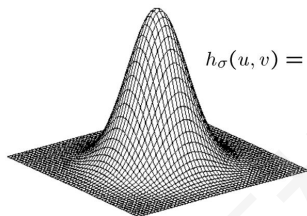
Laplacian of Gaussian ($\nabla^2 n_\sigma$ or $\nabla^2 G$)

$\nabla^2(n_\sigma * f) = \nabla^2(n_\sigma) * f$...saves us one operation.



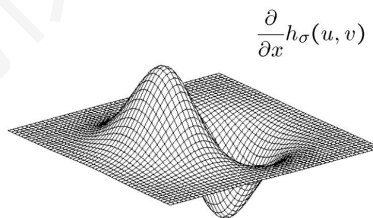
规格严格 功夫到家

对于二维信号



Gaussian

$$h_\sigma(u, v) = \frac{1}{2\pi\sigma^2} e^{-\frac{u^2+v^2}{2\sigma^2}}$$

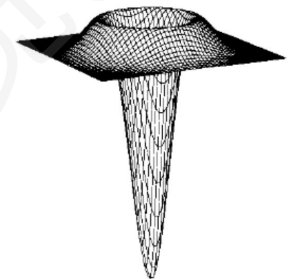


高斯导数滤波器(DoG)

$$\frac{\partial}{\partial x} h_\sigma(u, v)$$



$$\nabla^2 h_\sigma(u, v)$$



高斯拉普拉斯滤波器(LoG)

∇^2 is the Laplacian operator:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

高斯拉普拉斯滤波器(LoG)



高斯拉普拉斯滤波器

ML Center, HIT



拉普拉斯滤波器

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高斯拉普拉斯滤波器(LoG)



高斯拉普拉斯滤波器

ML Center, HIT

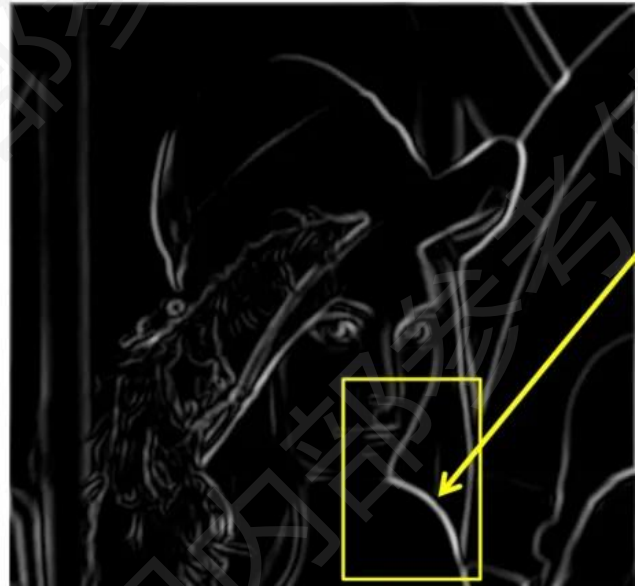
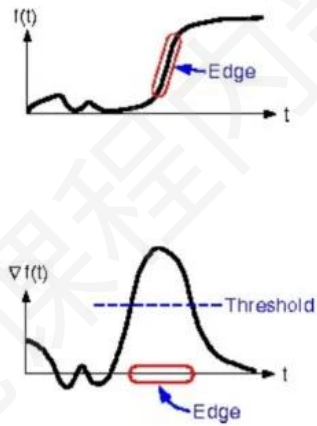


高斯导数滤波器

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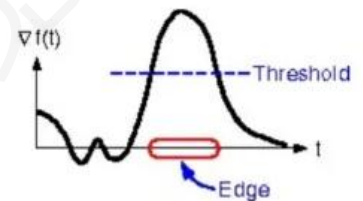
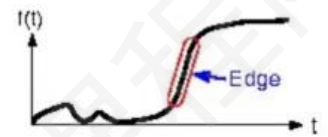
Problems



規格嚴格 功夫到家



Canny Operator



規格嚴格 功夫到家

Example



Demo: <http://bigwww.epfl.ch/demo/ip/demos/edgeDetector/>

Image from <http://bigwww.epfl.ch/demo/ip/demos/edgeDetector/>

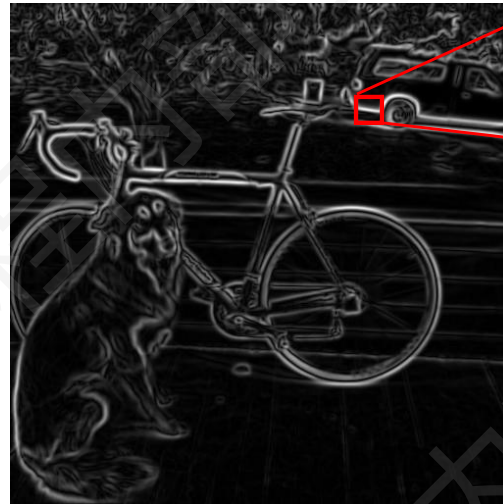
Finding edges



Smoothed gradient magnitude

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Finding edges



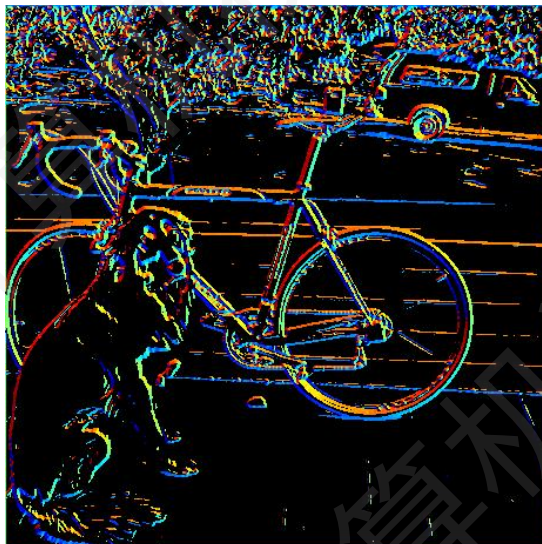
where is the edge?

Thresholding

規格嚴格 功夫到家

Get Orientation at Each Pixel

- Get orientation (below, threshold at minimum gradient magnitude)



$$\theta = \text{atan2}(g_y, g_x)$$

360

Gradient orientation angle

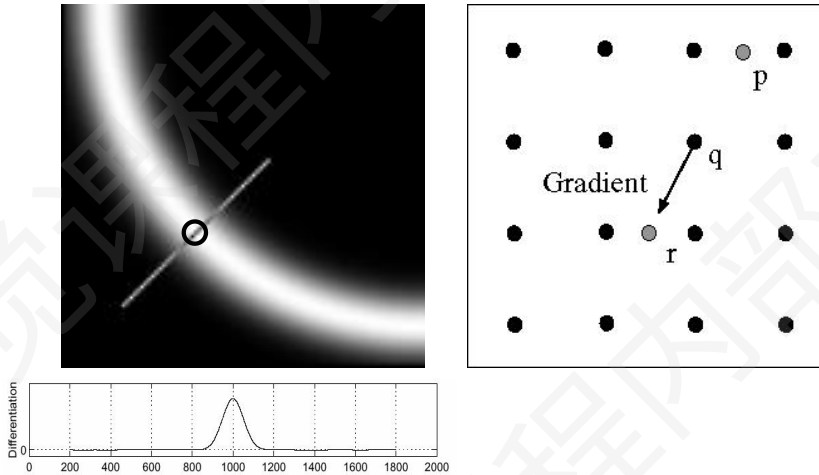
0

規格嚴格 功夫到家

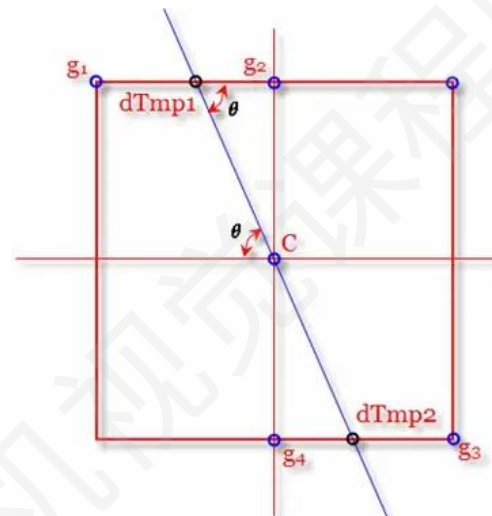
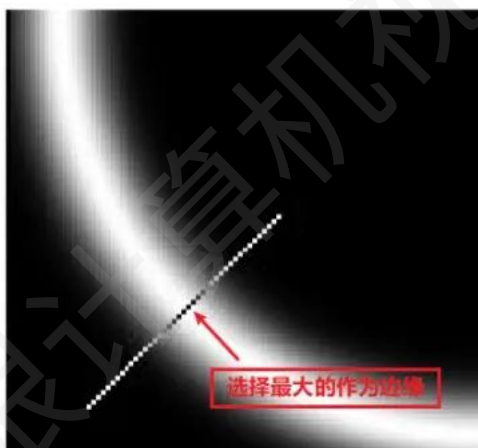


Non-maximum suppression

- Check if pixel is local maximum along gradient direction
- requires *interpolating* pixels p and r



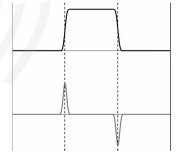
规格严格 功夫到家



规格严格 功夫到家

NMS (Non-maximum suppression)

$$n_{\sigma} * I \rightarrow \nabla n_{\sigma} * I \rightarrow \|\nabla n_{\sigma} * I\| \quad \hat{n} = \frac{\nabla n_{\sigma} * I}{\|\nabla n_{\sigma} * I\|} \rightarrow \frac{\partial^2 (n_{\sigma} * I)}{\partial \hat{n}^2}$$



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Before Non-max Suppression



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After Non-max Suppression



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Thresholding edges

- Still some noise
- Only want strong edges
- 2 thresholds, 3 cases
 - $R > T$: **strong edge**
 - $R < T$ but $R > t$: **weak edge**
 - $R < t$: **no edge**
- Why two thresholds?



规格严格 功夫到家

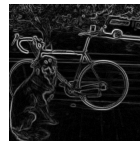
- Strong edges are edges!
- **Weak edges are edges only if they connect to strong**
- Look in some neighborhood (usually 8 closest)



规格严格 功夫到家

Canny edge detector

1. Filter image with derivative of Gaussian



2. Find magnitude and orientation of gradient



3. Non-maximum suppression

4. Linking and thresholding (hysteresis):

- Define two thresholds: low and high
- Use the high threshold to start edge curves and the low threshold to continue them



Source: D. Lowe, L. Fei-Fei, J. Sivic

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Canny edge detector

- Our first computer vision pipeline!
- Still a widely used edge detector in computer vision

J. Canny, [*A Computational Approach To Edge Detection*](#),
 IEEE Trans. Pattern Analysis and Machine Intelligence,
 8:679-714, 1986.

- Depends on several parameters:

high threshold

low threshold

σ : width of the Gaussian blur

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Canny edge detector



original

Canny with $\sigma = 1$

Canny with $\sigma = 2$

- The choice of σ depends on desired behavior
 - large σ detects “large-scale” edges
 - small σ detects fine edges

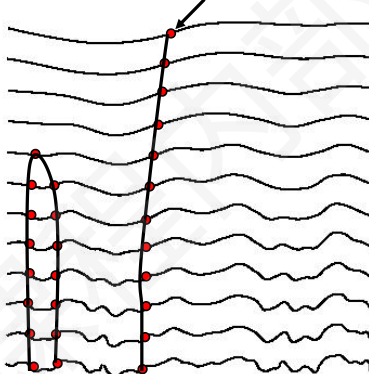
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Scale space [Witkin 83] first derivative peaks

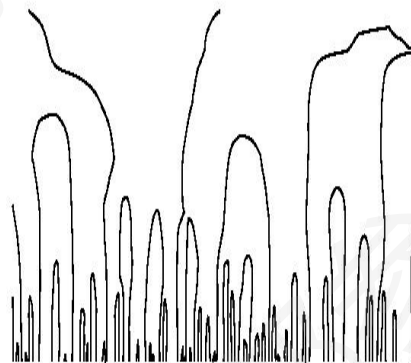
➤ Properties of scale space

- edge position may shift
- two edges may merge
- an edge may *not* split

larger σ



Gaussian filtered signal



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Canny边缘检测算法

Canny算法通常通过以下步骤实现：

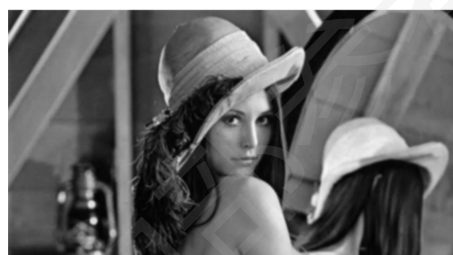
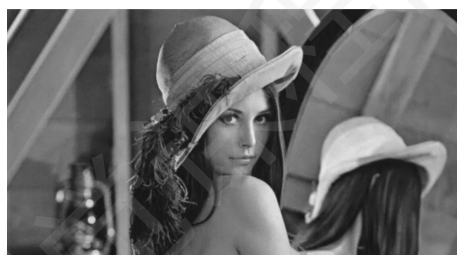
1. 图像平滑：即抑制噪声；
2. 图像梯度计算；
3. 边缘强度非最大值抑制（non-maximum suppression, NMS）；
4. 基于滞后的阈值化（hysteresis-based thresholding）的边缘定位。

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Canny边缘检测算法

图像平滑

由于边缘检测对图像噪声非常敏感，需要对图像进行去噪
使用高斯滤波器进行去噪



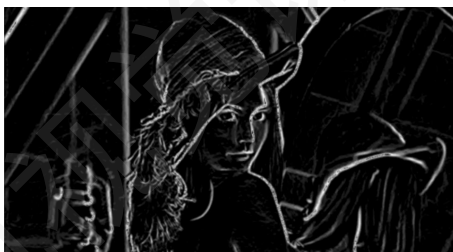
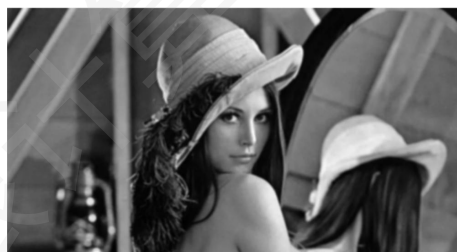
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Canny边缘检测算法

图像梯度计算

计算每个像素的梯度大小 $E_s(i, j | I) = \|\nabla I(i, j)\| = \sqrt{I_x^2(i, j) + I_y^2(i, j)}$

计算每个像素的梯度方向 $E_\theta(i, j | I) = \tan^{-1} \frac{I_y(i, j)}{I_x(i, j)} + \frac{\pi}{2}$



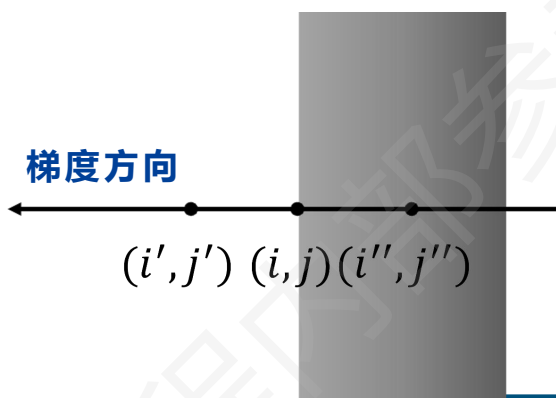
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Canny边缘检测算法

边缘强度非最大值抑制

非最大值抑制沿着每个梯度方向找到边缘强度局部最大的像素点予以保留
去除掉该方向上边缘强度非局部最大的像素点

对当前点 (i, j) ：若该点边缘强度在沿梯度方向的三个相邻点中最大，则保留该点在边缘中，反之将其从边缘中移除

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Canny边缘检测算法

边缘强度非最大值抑制

非最大值抑制沿着每个梯度方向找到边缘强度局部最大的像素点予以保留
去除掉该方向上边缘强度非局部最大的像素点

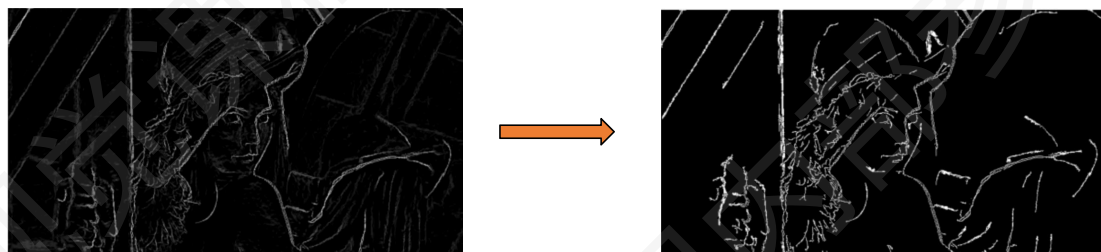
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Canny边缘检测算法

基于滞后的阈值化的边缘定位

设定两个阈值 T_h 和 T_l

- 若某一像素点边缘强度大于 T_h ，则认定其为（强）边缘点
- 若某一像素点边缘强度小于 T_l ，则认定其非边缘点
- 若某一像素点边缘强度介于 $T_l \sim T_h$ ，如果其处于强边缘点的邻域内，则认定其为（弱）边缘点



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Canny边缘检测算法

最终，Canny边缘检测算法的效果如下



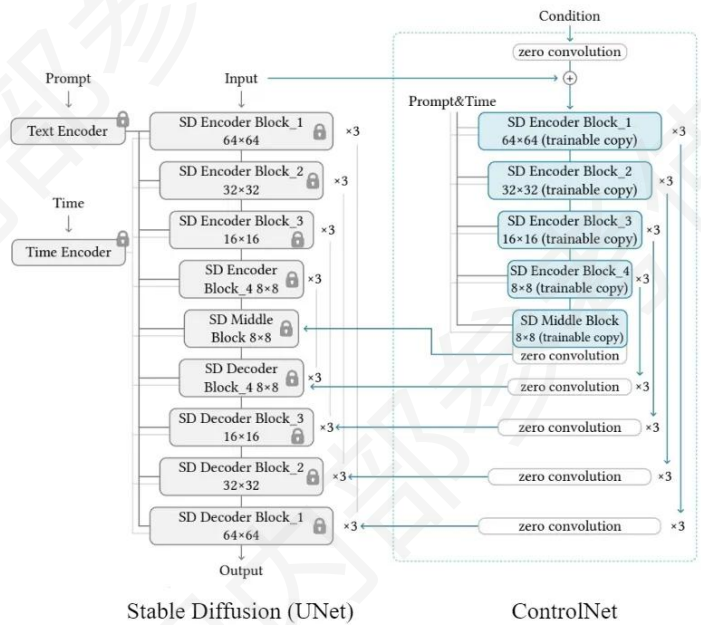
Kernel Size: 7, Sigma: 2

Kernel Size: 9, Sigma: 2.5

Kernel Size: 9, Sigma: 3

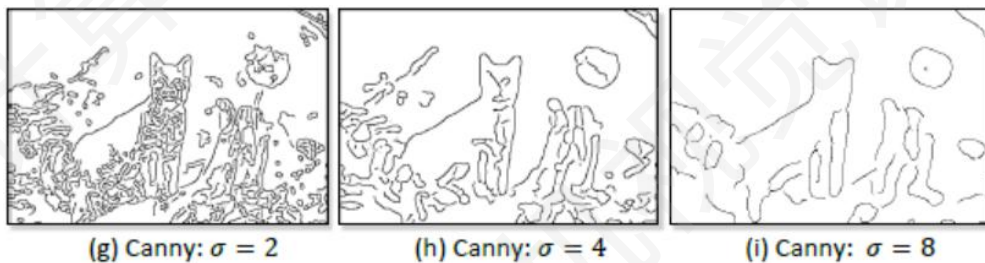


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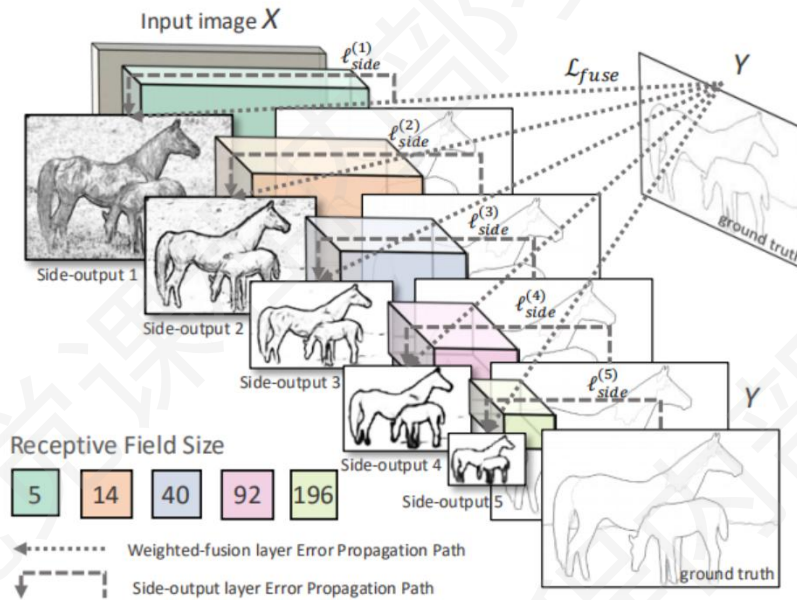
规格严格 功夫到家

Edge Detection with DL ?



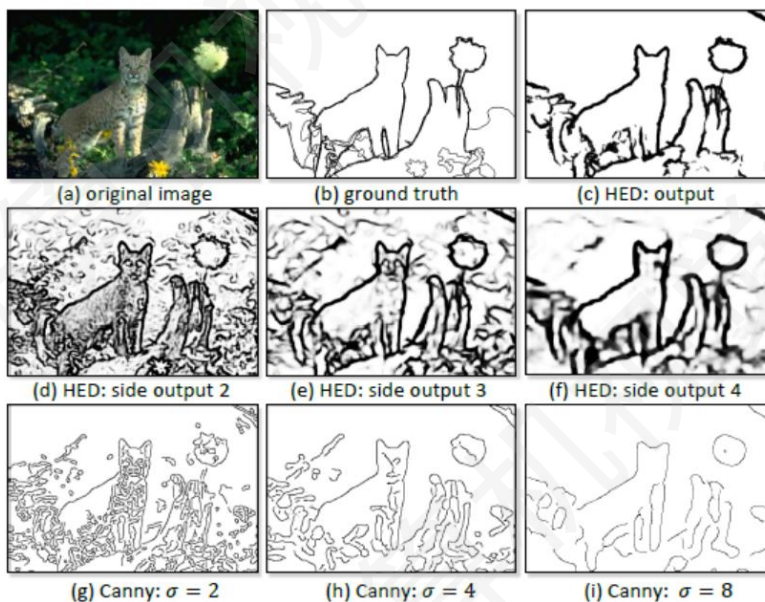
规格严格 功夫到家

Holistically-Nested Edge Detection (HED)



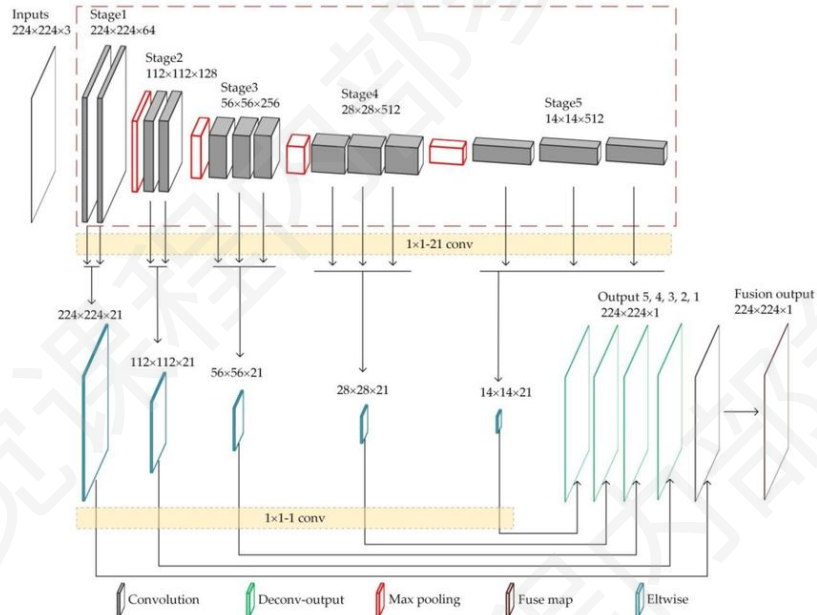
规格严格 功夫到家

Holistically-Nested Edge Detection (HED)



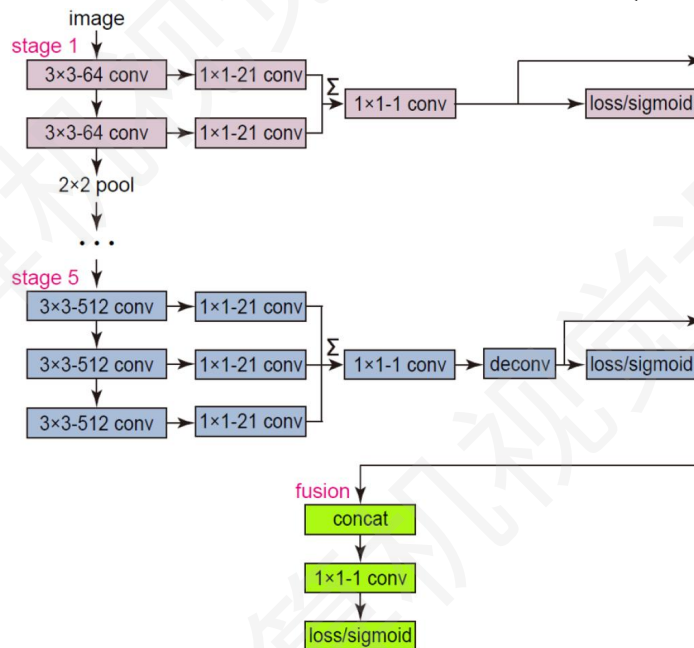
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Richer convolutional feature (RCF)



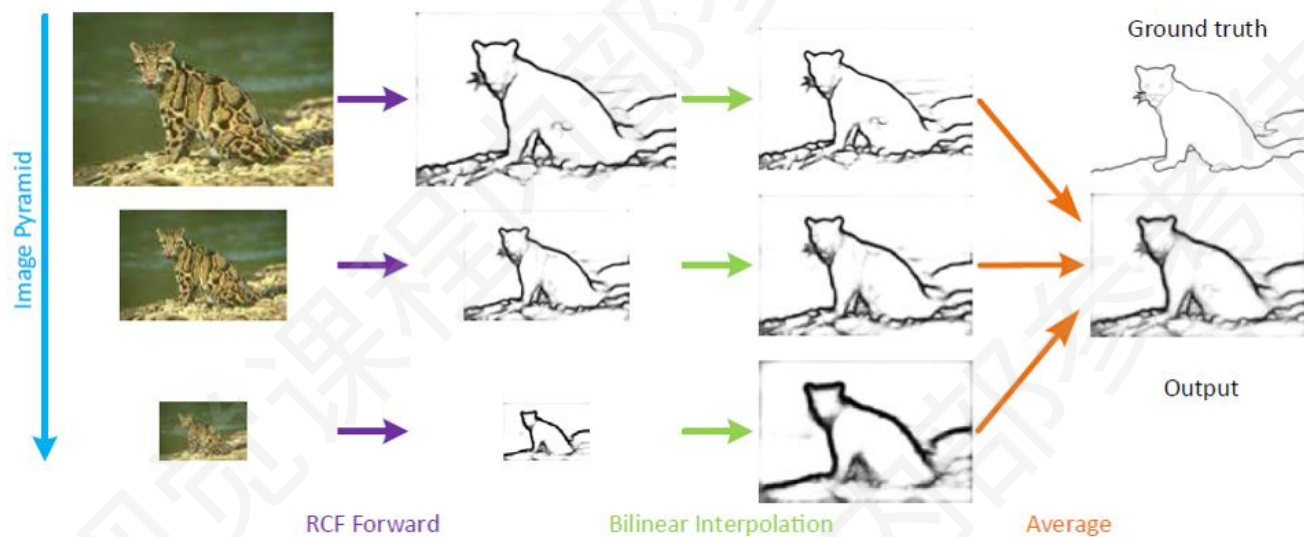
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Richer convolutional feature (RCF)



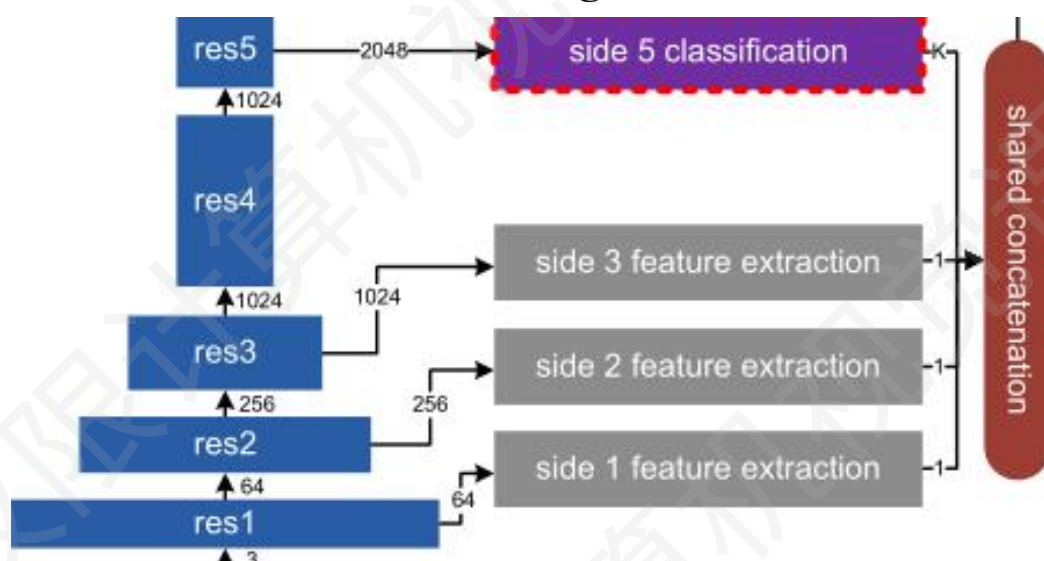
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Richer convolutional feature (RCF)



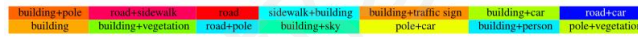
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Contour-Aware Semantic Segmentation Network

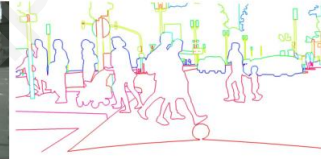


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Contour-Aware Semantic Segmentation Network



(a) Input image



(b) Ground truth



(c) CASNet output

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Summary of Sharpen Operators

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原始图像



-1	0
0	1

0	-1
1	0

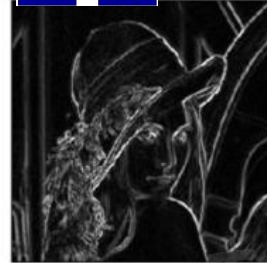
Prewitt算子



-1	-1	-1
0	0	0
1	1	1

-1	0	1
-1	0	1
-1	0	1

Sobel算子



-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

Sobel算子



Laplacian算子

$$\nabla^2 f = [f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1)] - 4f(x,y)$$



Canny算子



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